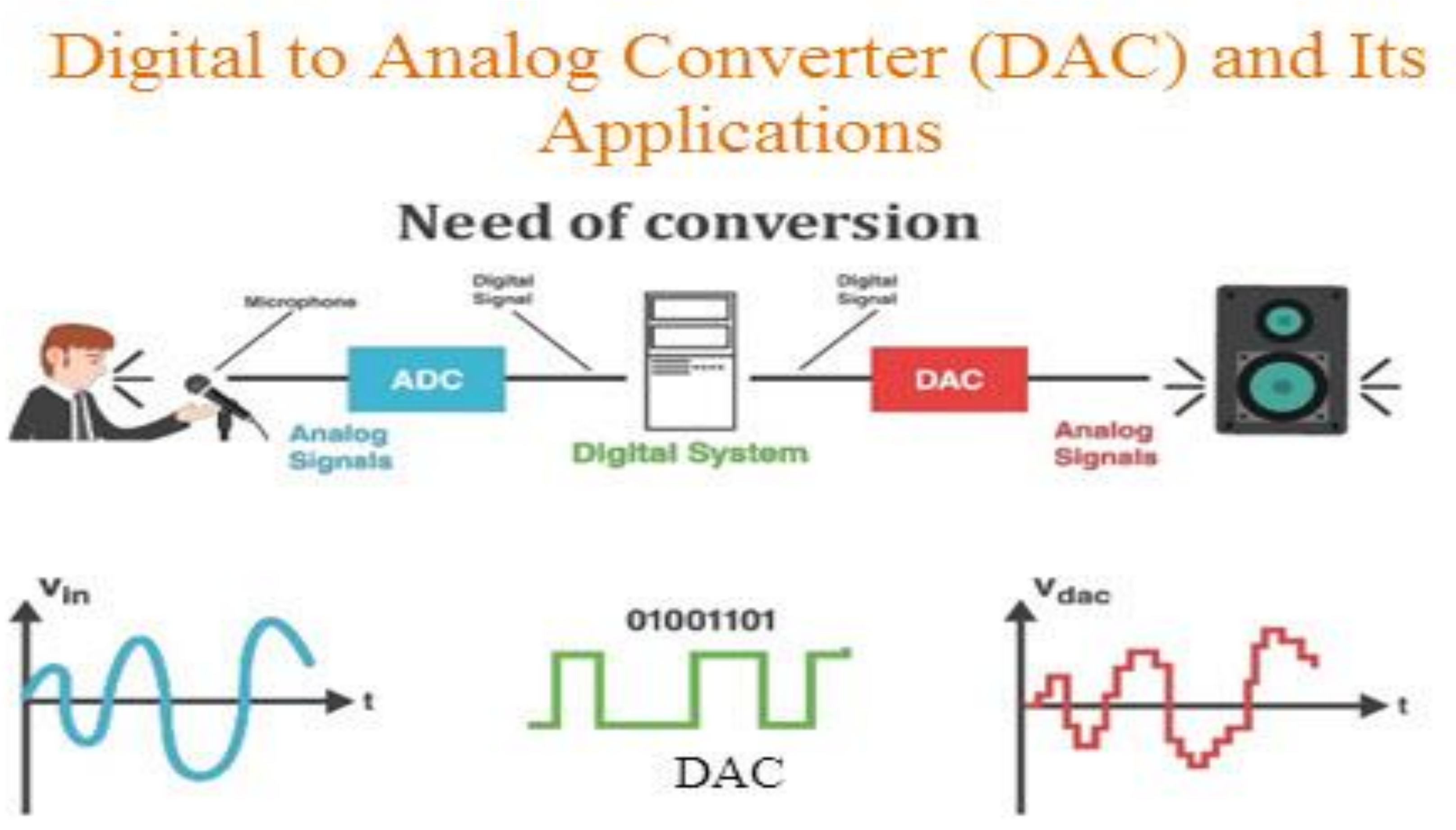
Module 5
Data Converters

Analog to Digital Converter (ADC)
and
Digital to Analog Converter (DAC)

Need of Digital systems

- Digital systems are more advantageous
- Processing signal using digital systems is need
- Due to advantages of digital systems, they are used in the following fields:
  - Control
  - Communication
  - Computers
  - Instrumentation, etc.
- So analog signal needs to be converted into digital and vice versa

Real life application of ADC and DAC



The output of the system may be required to be in the analog form and, therefore, the digital output has to be converted back to the analog form. The process is referred to as a *digital-to-analog conversion* and the system used for this purpose is referred to as a *digital-to-analog converter* (*D/A converter or DAC*). Some of the examples where A/D and D/A converters are used are:

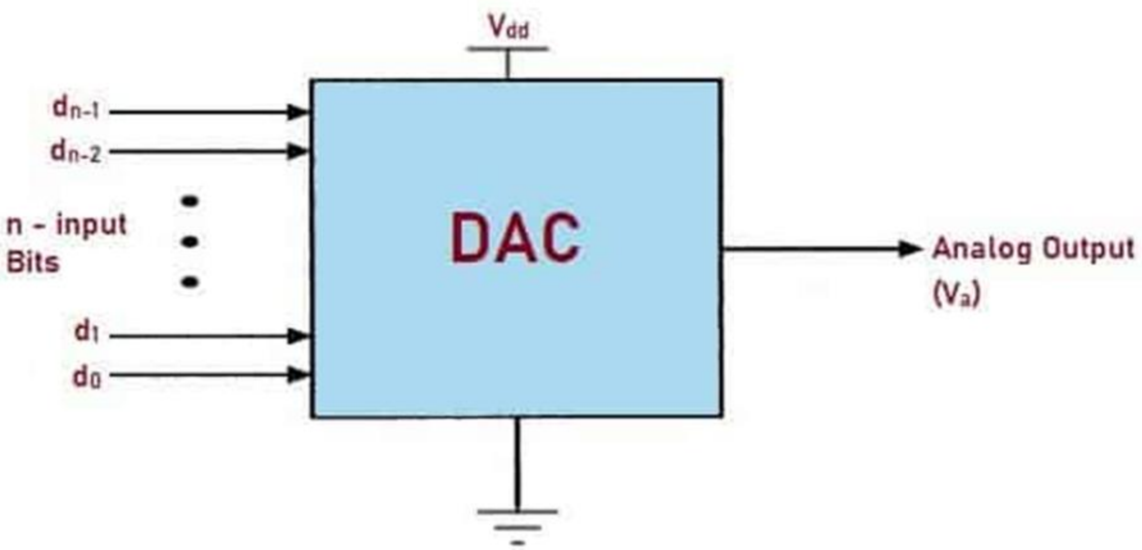
1. A digital system can be used to monitor the ambient temperature of an oven and if it exceeds a certain limit, it should reduce the fuel input. Here, an A/D converter is required to convert the output of the sensor (which converts temperature to an analog electrical signal) to digital form. If the temperature exceeds the specified limit, some digital output is produced which is to be converted to analog form in order to control the device which reduces the fuel input.

2. A digital voltmeter is used to measure an analog voltage and display the voltage in numerical form. In this, an A/D converter is required to convert the analog voltage into a digital signal. The required processing consists of determining its value. The output, in this case, is not required to be converted back to the analog form and hence a D/A converter is not needed.
3. A digital communication system is used to transmit messages which are in the form of analog electrical signals. This requires an A/D converter at the transmitting end and a D/A converter at the receiving end.

In microprocessor based process control systems, A/D and D/A converters are often used and are referred to as *peripherals* or I/O devices.

Since D/A converters are used as sub-systems in many A/D converters, D/A converters will be discussed first.

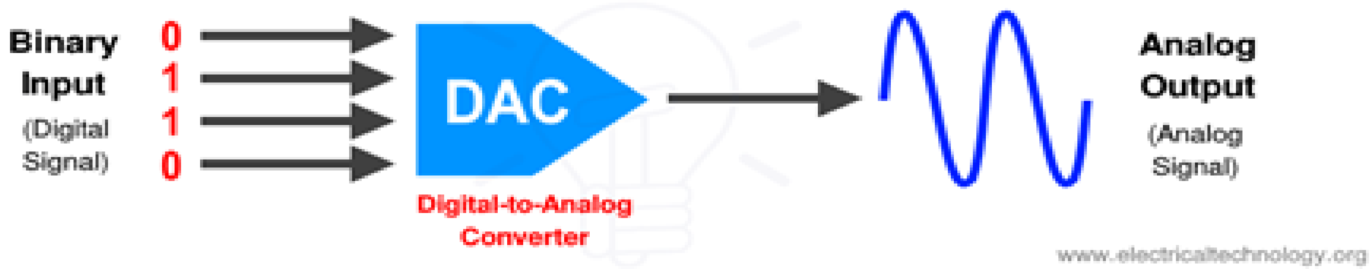
Basic block diagram of Digital to Analog Converter(DAC)



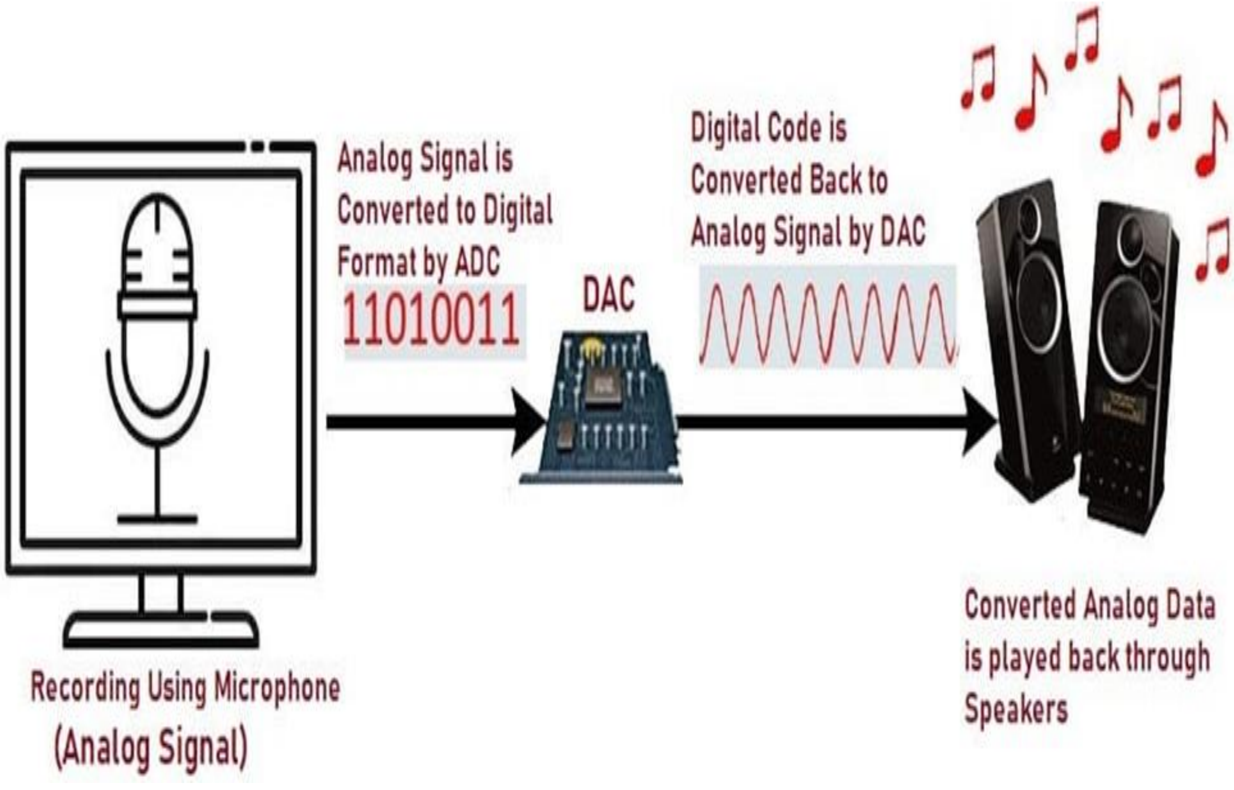
DAC

DIGITAL TO ANALOG CONVERTER

Types, Working, Block Diagram & Applications



Working Principle of DAC



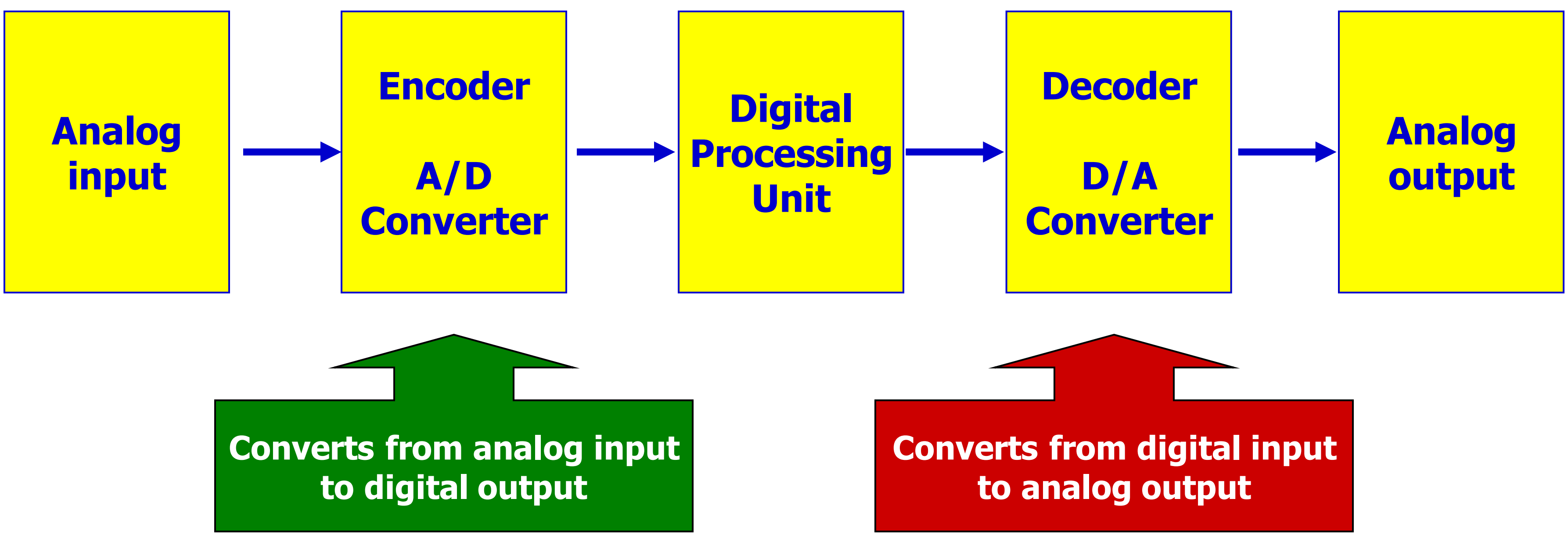
Applications of Digital to Analog Converter (DAC)

The applications of Digital to Analog Converter include:

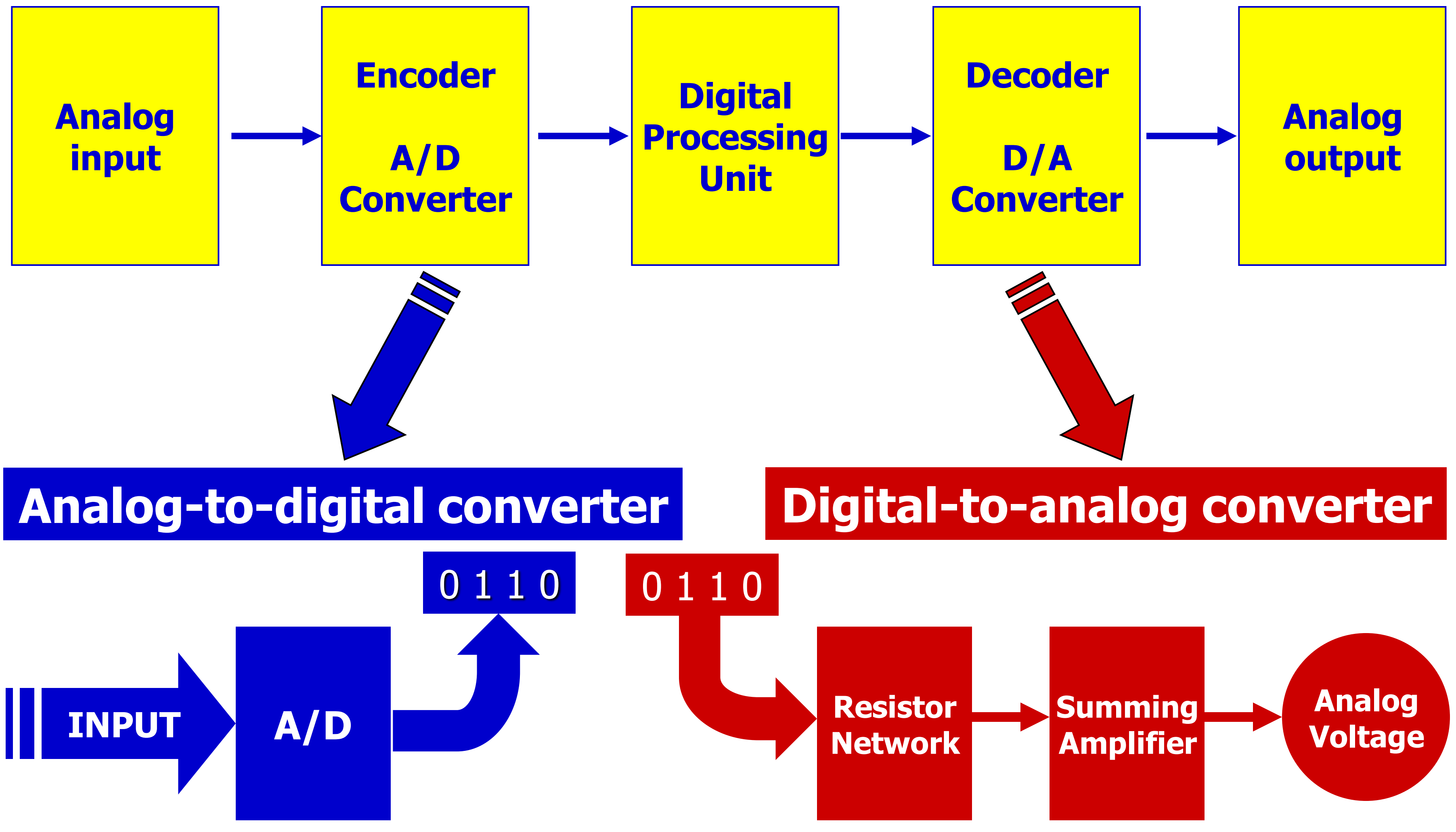
- DAC’s are used in Digital Signal Processing.
- They are also used in digital power supplies for Micro-controller.
- DAC’s are used in digital potentiometers.
- They are used in all digital data acquisition systems.

- **Hybrid System (analog-to-digital-to-analog)**
- **Simple 4-bit D/A Converter**
- **Simple D/A Converter Operation**
- **Counter-ramp-type A/D Converter**
- **Ramp-type A/D Converter**
- **Successive-approximation A/D Converter**
- **Flowcharting- Successive Approximation**
- **A/D Converter Specifications**

HYBRID SYSTEM
(BOTH ANALOG AND DIGITAL)



A/D AND D/A CONVERTERS



Basic principle of DAC

The input to a D/A converter is an *N*-bit binary signal, available in parallel form. Normally, digital signals are available at the output of the latches or registers and the voltages, corresponding to logic 0 and logic 1, available to drive the converter are in general not precisely fixed voltages. Therefore, these voltages are not applied directly to the converter but are used to operate digitally controlled switches. The switch is thrown to one of the two positions depending upon the digital signal (1 or 0) which connects precisely fixed voltages $V(1)$ or $V(0)$ to the converter input, corresponding to 1 or 0, respectively.

The analog output voltage V_0 of an *N*-bit straight binary D/A converter is related to the digital input by the equation

$$V_0 = K(2^{N-1} b_{N-1} + 2^{N-2} b_{N-2} + \ldots + 2^2 b_2 + 2b_1 + b_0) \qquad (10.1)$$

where K is a proportionality factor.

$$b_n = 1 \text{ if the } n\text{th bit of the digital input is 1}$$
$$= 0 \text{ if the } n\text{th bit of the digital input is 0}$$

10.2.1 Weighted-Resistor D/A Converter

Let us assume an N-bit straight binary input to a resistor network (through digitally controlled electronic switches) which produces a current I corresponding to logic 1 at the most-significant bit, $I/2$ corresponding to logic 1 at the next lower bit, $I/2^2$ for logic 1 at the next lower bit and so on, and $I/2^{N-1}$ for logic 1 at the least-significant bit position. The total current thus produced will be proportional to the digital input. This current can be converted into a corresponding voltage, by using an OP AMP, which will be proportional to the digital input.

The circuit of Fig. 10.1 can be used for converting the digital input to analog output which operates according to the above principle. This circuit is referred to as a *weighted-resistor D/A converter* since the resistance values are weighted in accordance with the binary weights.

In the circuit Fig. 10.1, the digital inputs (1 or 0) operate the switches. A switch is thrown to position 1 or 0 for a digital input corresponding to that bit being 1 or 0, respectively. The voltage applied to a resistor is $V(1)$ if the switch connected to it is in position 1 and $V(0)$ if it is in position 0. The current, I_i is given by

$$I_i = I_{N-1} + I_{N-2} + I_{N-3} + \dots + I_2 + I_1 + I_0 \tag{10.2}$$

where

$$\left.\begin{aligned} I_{N-1} &= V_{N-1}/R \\ I_{N-2} &= V_{N-2}/2R \\ I_{N-3} &= V_{N-3}/2^2R \\ &\vdots \\ I_0 &= V_0/2^{N-1}R \end{aligned}\right\} \text{where } \begin{aligned} V_n &= V(1) \text{ if } b_n = 1 \\ &= V(0) \text{ if } b_n = 0 \end{aligned} \tag{10.3}$$

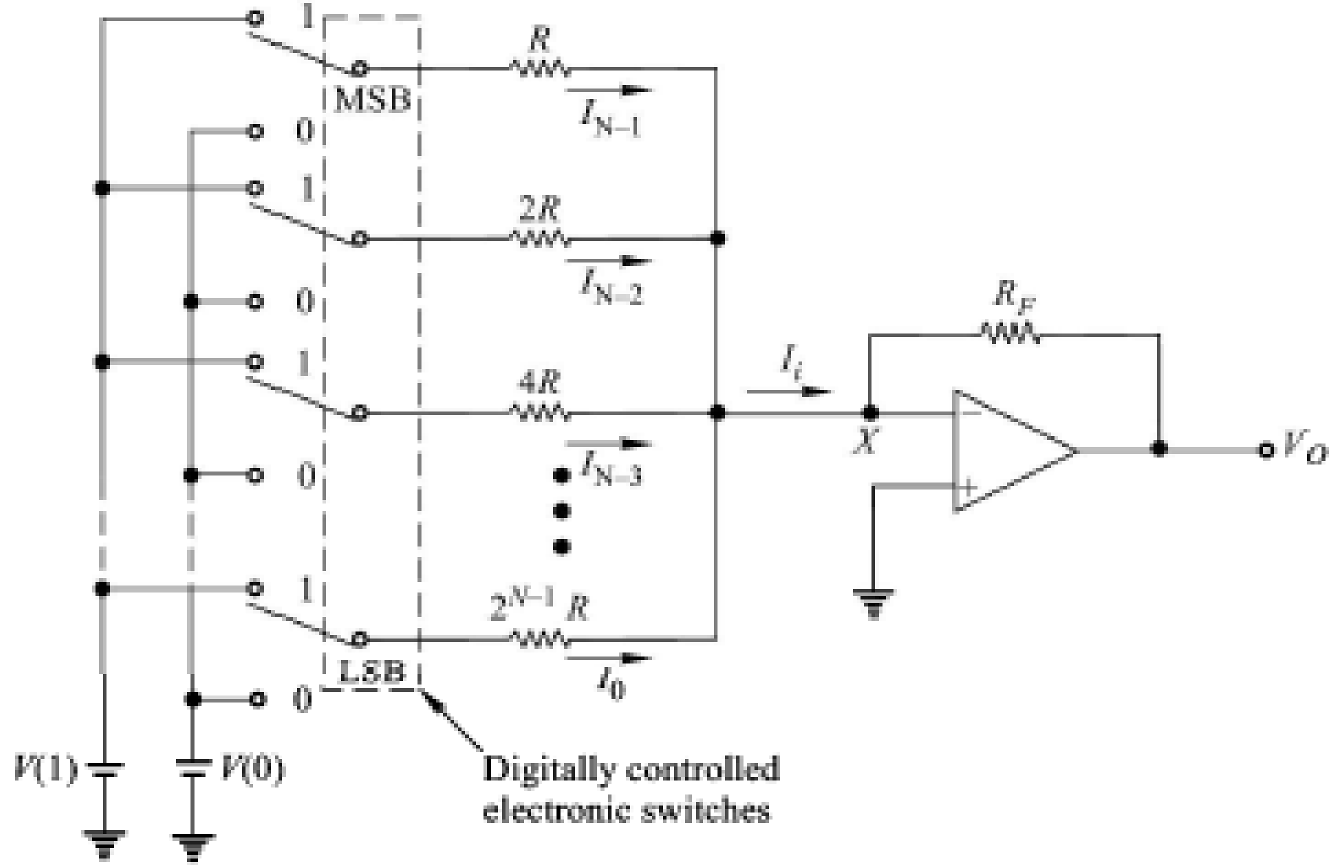


Fig. 10.1 ***Weighted-Resistor D/A Converter***

For straight binary input, $V(0) = 0$ and $V(1) = -V_R$, and the output voltage V_O is given by

$$V_O = -(-V_R)\left(\frac{R_F}{R}b_{N-1} + \frac{R_F}{2R}b_{N-2} + \frac{R_F}{2^2 R}b_{N-3} + \ldots + \frac{R_F}{2^{N-1}R}b_0\right) \tag{10.4}$$

which is of the same form as Eq. (10.1)

with
$$K = \frac{R_F}{2^{N-1}R} \cdot V_R \tag{10.5}$$

The output swings in only one direction and therefore is *unipolar*. If it is required to convert digital data in bipolar format such as in sign-magnitude, 1's complement or 2's complement format, then $V(0) \neq 0$. In such cases, $V(0)$ is used to *offset* the output swing.

With $V(1)$ and $V(0)$ as the voltages applied to the resistor network for 1 and 0, respectively, the output voltage V_O of Fig. 10.1 is given by

$$V_O = \frac{R_F}{2^{N-1}R}\left(2^{N-1}V_{N-1} + 2^{N-2}V_{N-2} + \ldots + 2^1 V_1 + 2^0 V_0\right) \tag{10.6}$$

An offset can also be produced in the output voltage V_O by using the circuit of Fig. 10.2. The offset voltage produced in this circuit is

$$-\frac{R_F}{R_{\text{off}}} \cdot V_{\text{off}}$$

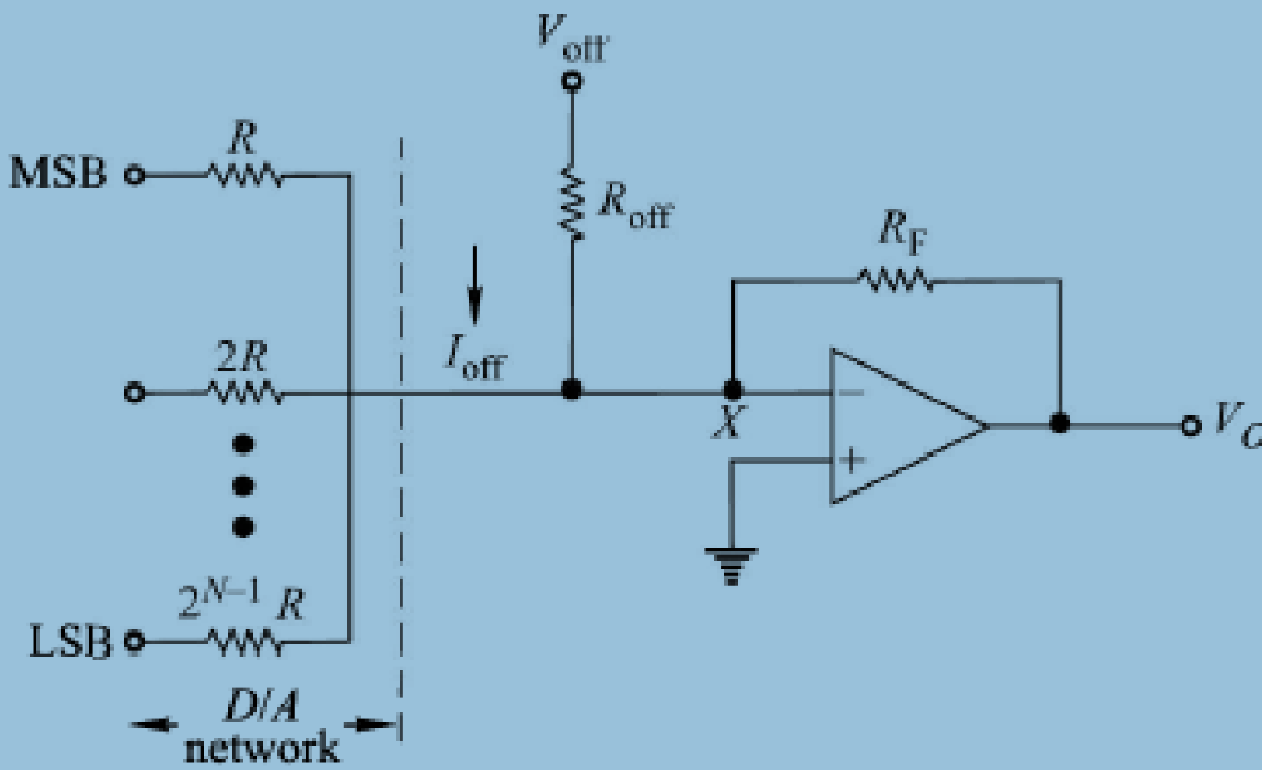


Fig. 10.2 ***Circuit Used to Offset the Output Voltage***

Drawbacks of weighted resistor DAC

The weighted-resistor D/A converter has the problem of having a wide range of resistor values (R to $2^{N-1} \cdot R$) with required precision and which track over a wide temperature range. It is difficult to fabricate such a wide range of resistance values in a monolithic IC. This difficulty is eliminated by R–$2R$ ladder D/A converter discussed below.

10.2.2 R–$2R$ Ladder D/A Converter

An R–$2R$ ladder D/A converter is shown in Fig. 10.3. It uses resistors of only two values, R and $2R$. The inputs to the resistor network are applied through digitally controlled switches. A switch is in position 0 or 1 corresponding to the digital input for that bit position being 0 or 1, respectively. To analyse this circuit, for simplicity we consider a 3-bit R–$2R$ ladder D/A network shown in Fig. 10.4. In this circuit, we have assumed the digital input as 001.

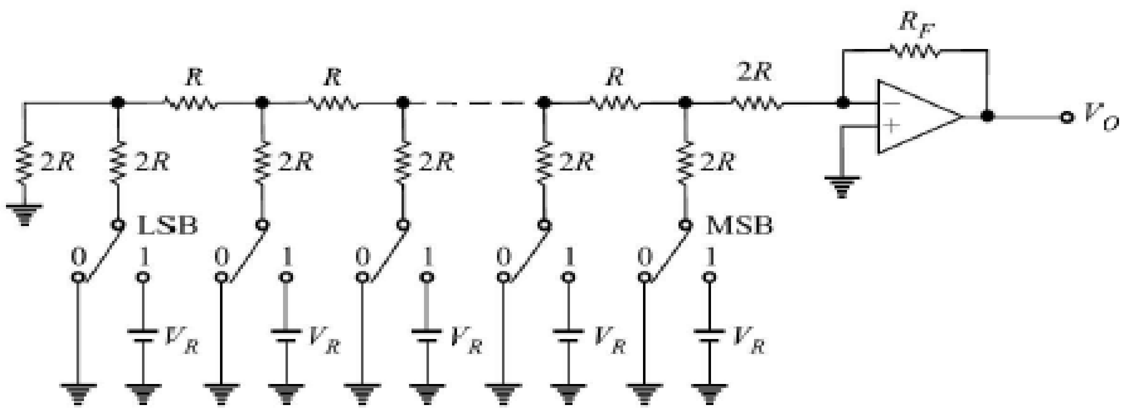


Fig. 10.3 ***R–2R Ladder D/A Converter***

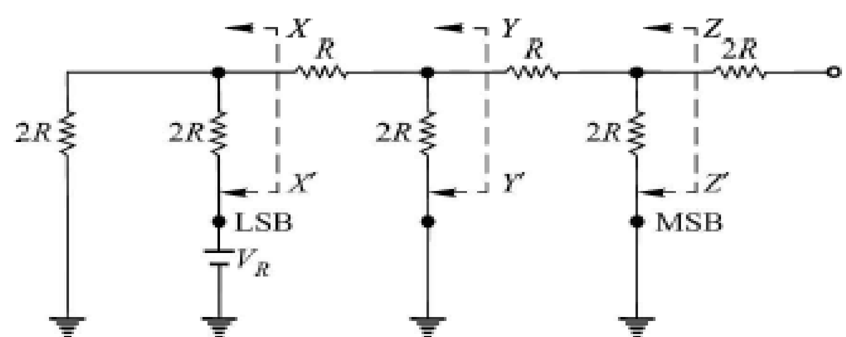


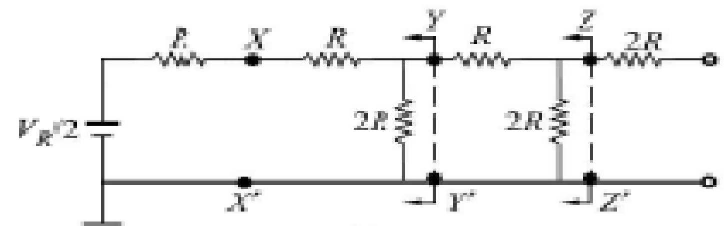
Fig. 10.4 ***3-bit R–2R Ladder D/A Network***

The circuit is simplified using Thevenin's theorem. Applying Thevenin's theorem at XX', we obtain the circuit of Fig. 10.5*a*. Similarly, applying Thevenin's theorem at YY' and ZZ', we obtain the circuits of Fig. 10.5*b* and *c*, respectively. Here, LSB has been assumed as 1 and the equivalent voltage obtained is $V_R/2^3$.

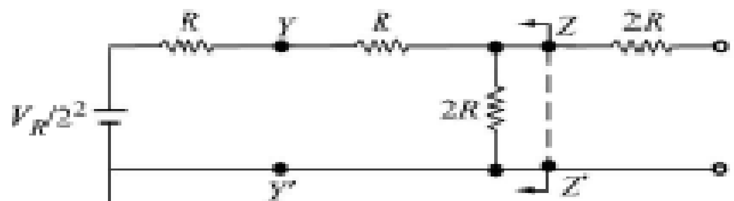
Similarly, for the digital input of 010 and 100, the equivalent voltages are $V_R/2^2$ and $V_R/2^1$, respectively. The value of the equivalent resistance is $3R$ in each case. Therefore, we obtain an equivalent circuit of Fig. 10.4 which is given in Fig. 10.5*d*. For the circuit of Fig. 10.5*d*, the output analog voltage V_O is given by

$$V_O = -\left(\frac{R_F}{3R}\cdot\frac{V_R}{2^3}b_0 + \frac{R_F}{3R}\cdot\frac{V_R}{2^2}b_1 + \frac{R_F}{3R}\cdot\frac{V_R}{2^1}b_2\right)$$

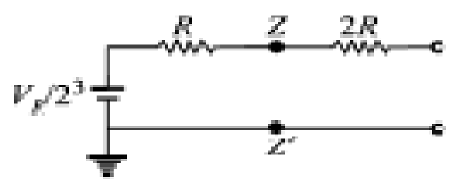
$$= -\left(\frac{R_F}{3R}\right)\cdot\left(\frac{V_R}{2^3}\right)[4b_2 + 2b_1 + 1b_0] \tag{10.7}$$



(a)



(b)



(c)

Equation (10.7) shows that the analog output voltage is proportional to the digital input. In general, for an N-bit D/A converter, the output voltage can be similarly determined and is given by

$$V_O = (2^{N-1} b_{N-1} + 2^{N-2} b_{N-2} + \ldots + 2^2 b_2 + 2^1 b_1 + 2^0 b_0) \tag{10.8}$$

where $R_F = 3R$ and $V_R = -2^N$ V

The number of resistors required for an N-bit D/A converter is $2N$ in the case of R–$2R$ ladder D/A converter, whereas it is only N in the case of a weighted-resistor D/A converter. But, because of the wide spread in the resistance values for large N, the weighted-resistor D/A converter is not suitable. However, the weighted-resistor network of Fig. 10.2 can be modified to accommodate a large number of bits without consequent spread in resistor values. One such circuit is shown in Fig. 10.6. Here, the bits are divided into groups of four. The most-significant four bits are applied in a manner similar to the one used in weighted-resistor D/A converter. The least-significant four bits are applied through an additional resistor r, in addition to weighted-resistors. This is done to produce input currents of OP AMP due to least-significant group of 4-bits and the most-significant group of 4-bits in the ratio of 1: 16 $\left(\frac{b_3}{b_7} = \frac{b_2}{b_6} = \frac{b_1}{b_5} = \frac{b_0}{b_4} = 1/16\right)$.

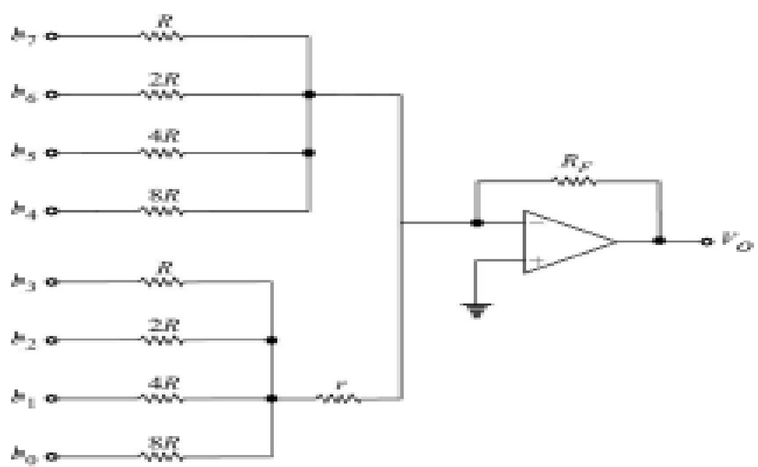


Fig. 10.6 ***Modified Weighted-Resistor D/A Converter***

The resistor r is determined in the following way:

Let the b_3 bit be 1 and b_2, b_1, and b_0 bits be all 0. The portion of the circuit corresponding to this is shown in Fig. 10.7a and its simplified version is shown in Fig. 10.7b.

From Fig. 10.7b, we obtain I_{in} assuming virtual short at the input of the OP AMP.

$$I_{in} = \frac{V_R}{R + \dfrac{r(8/7R)}{(r + 8/7R)}} \times \frac{\left(\dfrac{8}{7}R\right)}{\left(r + \dfrac{8}{7}R\right)} \tag{10.9}$$

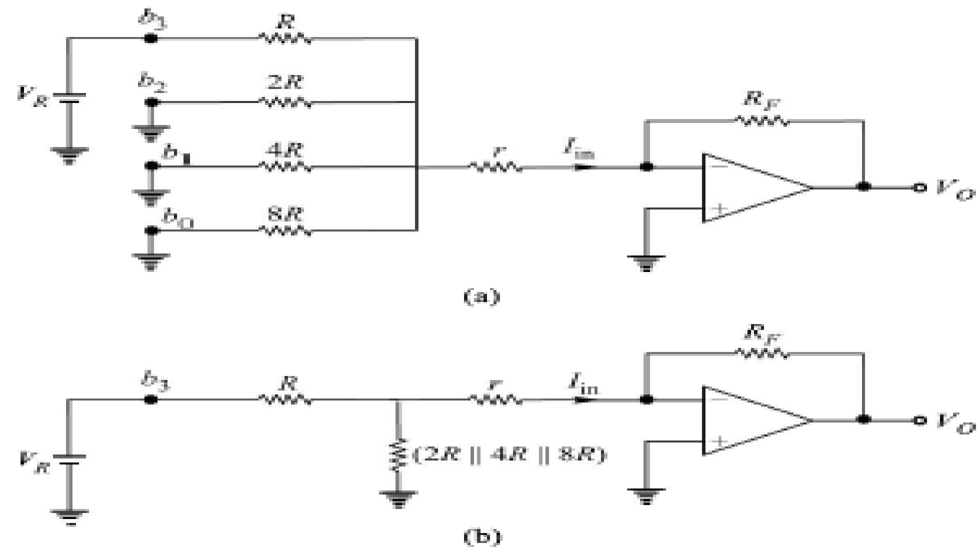


Fig. 10.7 ***A Portion of Modified Weighted-Resistor D/A Converter***

This current must be 1/16th of the current due to b_7, which is V_R/R. Therefore,

$$\frac{V_R\left(\frac{8}{7}R\right)}{R\left(r+\frac{8}{7}R\right)+r\left(\frac{8}{7}R\right)} = \frac{V_R}{16R}$$

or

$$r = 8R \tag{10.10}$$

With this value of r, it can be verified that the currents due to b_2, b_1, and b_0 will be 1/16th of the currents due to b_6, b_5, and b_4, respectively (Problem 10.3).

The output analog voltage V_O of the modified D/A converter of Fig. 10.6 for $r = 8R$ is given by

$$\begin{aligned} V_O &= -\left(\frac{V_R}{R}R_F b_7 + \frac{V_R}{2R}R_F b_6 + \frac{V_R}{4R}R_F b_5 + \frac{V_R}{8R}R_F b_4 + \frac{V_R}{16R}R_F b_3 \right.\\ &\qquad \left. + \frac{V_R}{32R}R_F b_2 + \frac{V_R}{64R}R_F b_1 + \frac{V_R}{128R}R_F b_0\right)\\ &= -\left(\frac{V_R}{2^7}\cdot\frac{R_F}{R}\right)\cdot(2^7 b_7 + 2^6 b_6 + \ldots + 2^2 b_2 + 2^1 b_1 + b_0) \end{aligned} \tag{10.11}$$

We observe from Eq. (10.11) that the analog output voltage is proportional to the digital input. The number of resistors in this circuit is less and also the spread in the resistor values is reduced. This configuration can be used for any number of bits.

Example

(a) Consider a 4-bit unipolar D/A converter with $V(1) = -1$ V, $V(0) = 0$ V, and $R_F = 8R$. Obtain the analog output voltage for each of the digital inputs from 0000 to 1111. (b) Adjust the offset voltage, using the circuit of Fig. 10.2, such that $V_O = 0$ V for a digital input of 1000. With this offset, obtain the analog output voltage for each of the digital inputs. (c) With the offset used above, obtain the analog output voltages if the MSB is complemented before applying to the D/A converter.

Using following equation we get the values as shown in the table

$$V_O = \frac{R_F}{2^{N-1} R} (2^{N-1} V_{N-1} + 2^{N-2} V_{N-2} + \ldots + 2^1 V_1 + 2^0 V_0) \quad (10.6)$$

Digital input				Analog output
b_3	b_2	b_1	b_0	V
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3

(Continued)

Table 10.1 (Continued)

Digital input				Analog output
b_3	b_2	b_1	b_0	V
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
1	0	0	0	8
1	0	0	1	9
1	0	1	0	10
1	0	1	1	11
1	1	0	0	12
1	1	0	1	13
1	1	1	0	14
1	1	1	1	15

Solution

(a) Using Eq. (10.6), we obtain the analog voltages which are same as those given in Table10.1.

(b) Without offset, the output voltage V_o is 8 V for the digital input 1000. Therefore, the offset must produce a voltage of – 8 V at the output, i.e.

$$-\frac{R_F}{R_{\text{off}}} V_{\text{off}} = -8 \text{ V}$$

We can use $R_{\text{off}} = R$ and $V_{\text{off}} = 1$ V. The analog voltages are given in Table 10.2.

(c) If the digital input to the D/A converter is $\bar{b}_3\, b_2\, b_1\, b_0$, then the analog output voltages can be obtained from Table 10.2, and these are given in Table 10.3.

From Table 10.3, we observe that this is a D/A converter which converts 2's complement format to analog signal.

Table 10.2

Digital input				Analog output
b_3	b_2	b_1	b_0	V
0	0	0	0	– 8
0	0	0	1	– 7
0	0	1	0	– 6
0	0	1	1	– 5

Digital input				Analog output
b_3	b_2	b_1	b_0	V
0	1	0	0	− 4
0	1	0	1	− 3
0	1	1	0	− 2
0	1	1	1	− 1
1	0	0	0	0
1	0	0	1	+ 1
1	0	1	0	+ 2
1	0	1	1	+ 3
1	1	0	0	+ 4
1	1	0	1	+ 5
1	1	1	0	+ 6
1	1	1	1	+ 7

Digital input				Analog output
b_3	b_2	b_1	b_0	V
1	0	0	0	−8
1	0	0	1	−7
1	0	1	0	−6
1	0	1	1	−5
1	1	0	0	−4
1	1	0	1	−3
1	1	1	0	−2
1	1	1	1	−1
0	0	0	0	0
0	0	0	1	+1
0	0	1	0	+2
0	0	1	1	+3
0	1	0	0	+4
0	1	0	1	+5
0	1	1	0	+6
0	1	1	1	+7

Example 10.4

Design a 2-decade BCD D/A converter.

Solution

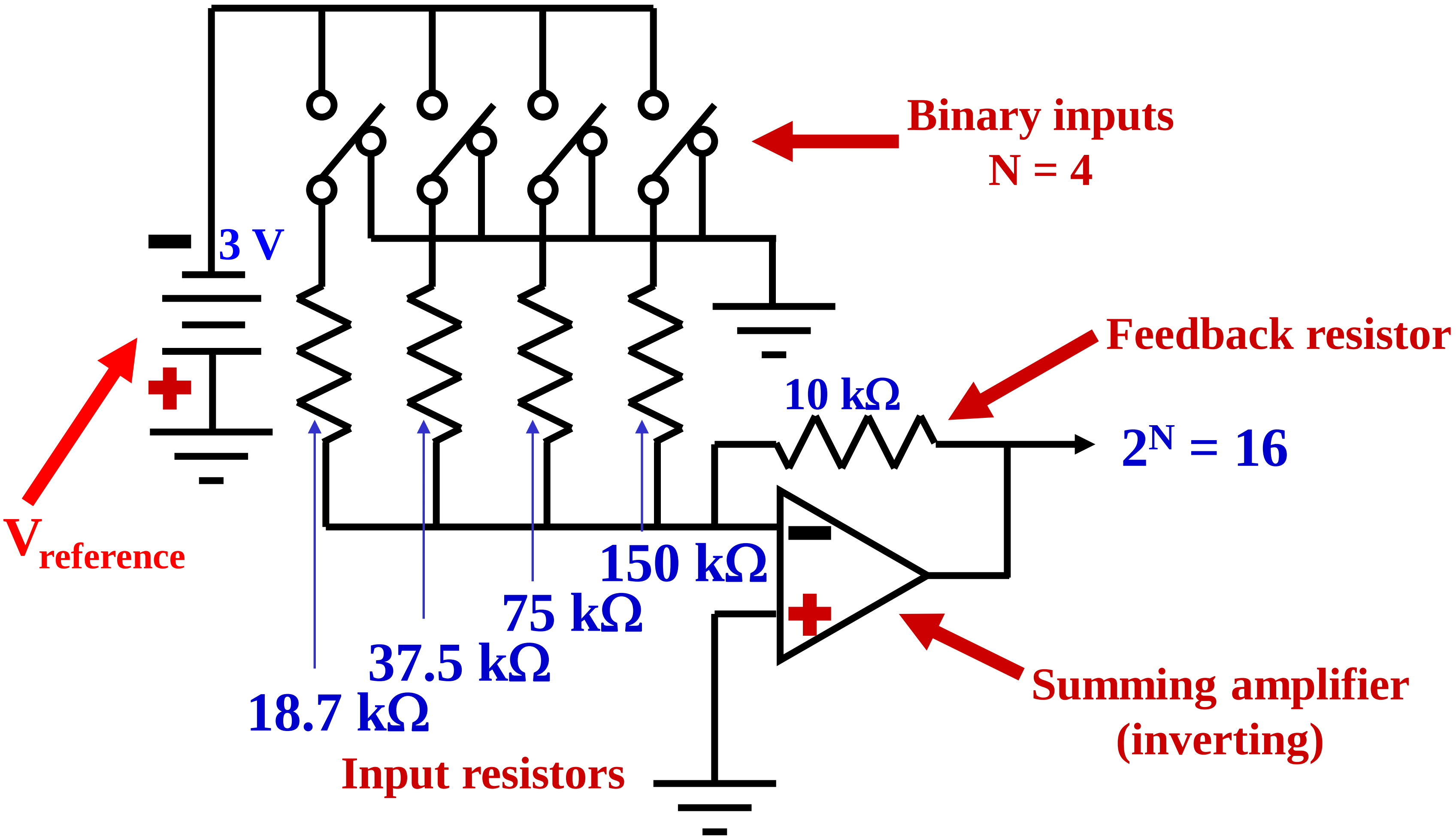
The circuit of Fig. 10.6 can be used for BCD D/A converter. The binary inputs corresponding to the least-significant digit are applied at b_3, b_2, b_1, and b_0 and those corresponding to the next digit, at b_7, b_6, b_5, and b_4. The value of r is chosen so as to make the input current of OP AMP corresponding to LSD as 1/10th of that of current due to MSD, and is given by

$$\frac{V_R\left(\frac{8}{7}R\right)}{R\left(r+\frac{8}{7}R\right)+r\left(\frac{8}{7}R\right)} = \frac{V_R}{10R}$$

or

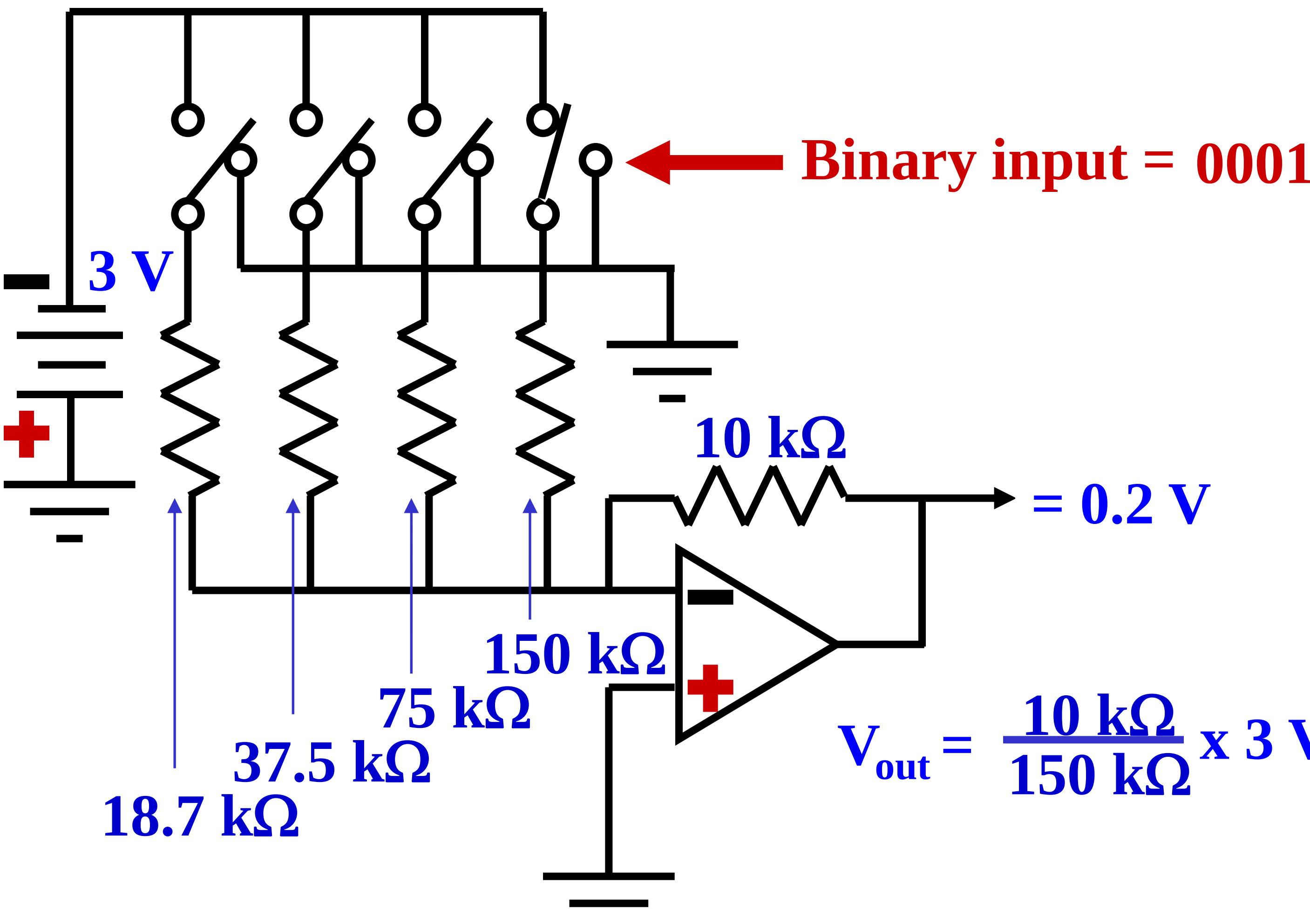
$$r = 4.8R \tag{10.12}$$

SIMPLE D/A CONVERTER



This circuit can produce 16 different analog output voltages.

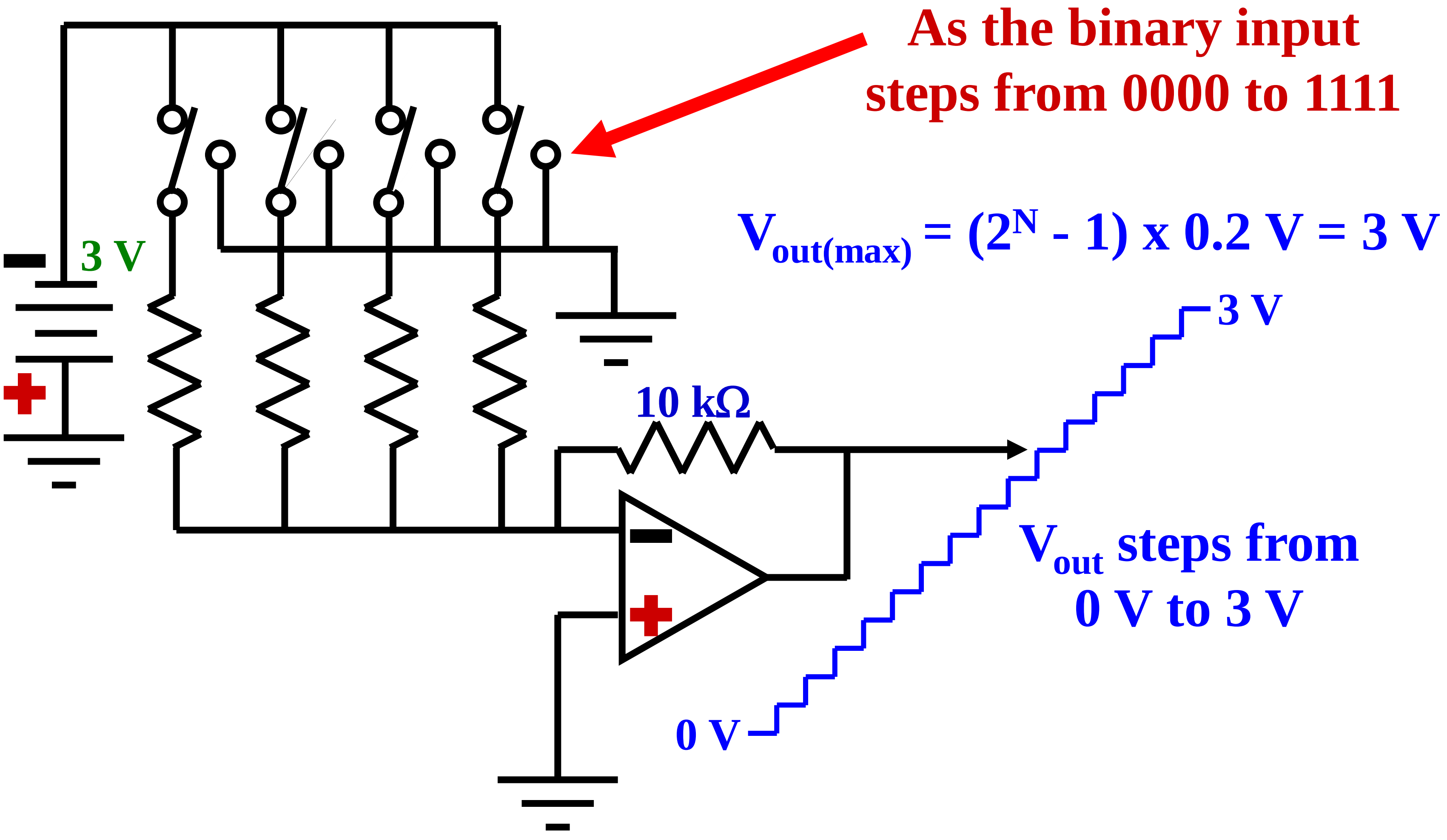
D/A CONVERTER OPERATION



$$V_{out} = \frac{10\ \text{k}\Omega}{150\ \text{k}\Omega} \times 3\ \text{V} = 0.2\ \text{V}$$

The smallest step size for this circuit is 0.2 V.

4-BIT D/A CONVERTER



The total span for this circuit is from 0 to 3 V.

Specifications of DAC

10.2.3 Specifications for D/A Converters

The characteristics of a D/A converter, which are generally specified by the manufacturers are:

1. Resolution,
2. Linearity,
3. Accuracy,
4. Settling time, and
5. Temperature sensitivity.

Analog to Digital Converters

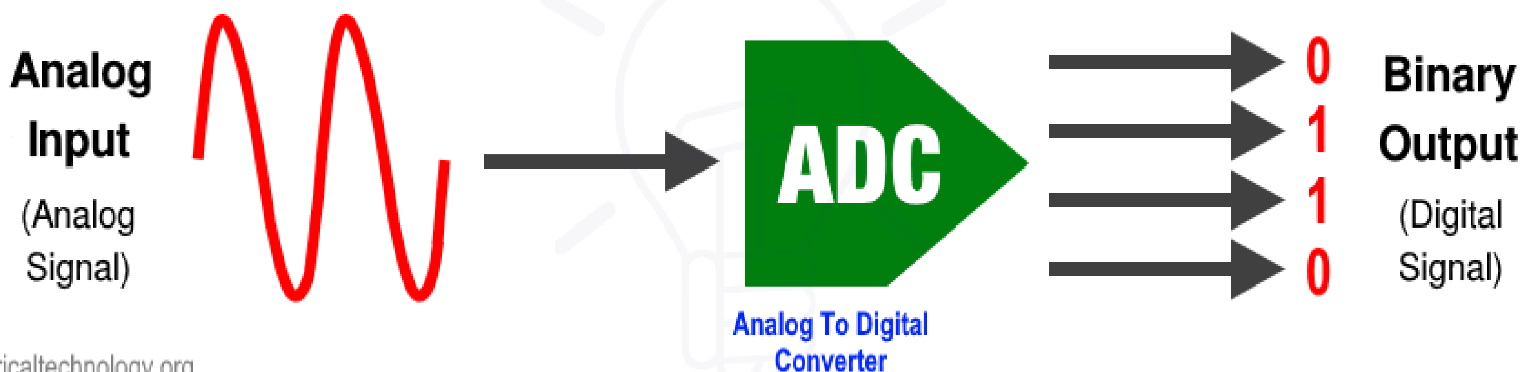
Introduction of ADC

- **What is ADC(Analog to Digital Converter)**
- **Why ADC is needed**
- **Application of ADC**
- **A/D conversion process**

ADC

ANALOG TO DIGITAL CONVERTER

Types, Working, Block Diagram & Applications



What is ADC

- An electronic circuit which transforms a signal from analog (continuous) to digital (discrete) form.
- Analog signals are directly measurable quantities.
- Digital signals only have two states. For digital computer, we refer to binary states, 0 and 1.

Need of ADCs

- Digital systems are more advantageous
- Processing signal using digital systems is need
- Due to advantages of digital systems, they are used in the following fields:
  - Control
  - Communication
  - Computers
  - Instrumentation, etc.

Why ADC is needed

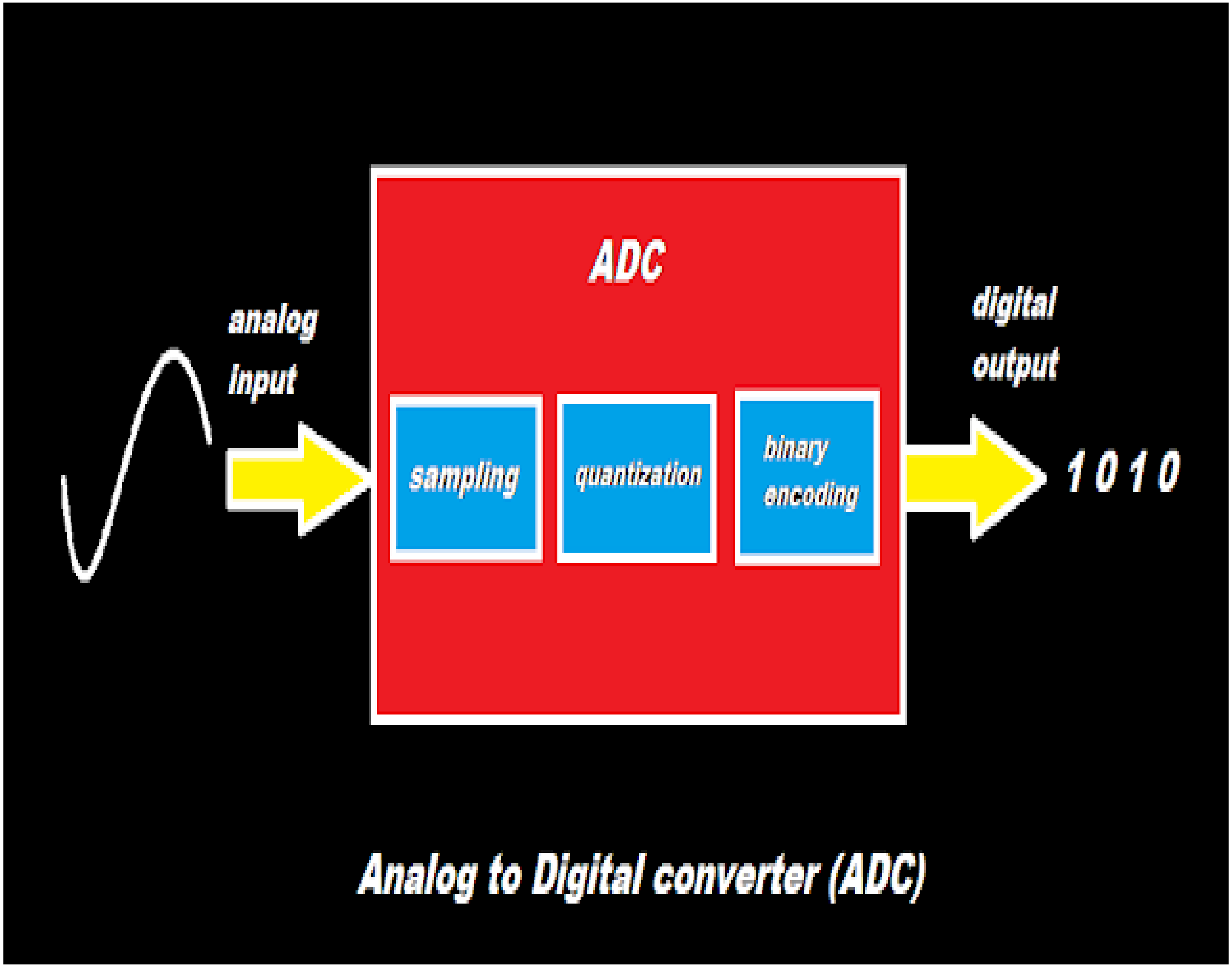
- **Microprocessors can only perform complex processing on digitized signals.**
- **When signals are in digital form they are less susceptible to the deleterious effects of additive noise.**
- **ADC Provides a link between the analog world of transducers and the digital world of signal processing and data handling.**

Processes in ADC

- Process of converting analog signal to digital form involves sequence of following processes:
  - Sampling
  - Holding
  - Quantizing
  - Encoding
- Sampling and holding are done simultaneously by Sample and Hold circuit
- Quantizing and encoding are done simultaneously by circuit known as Analog to digital converter

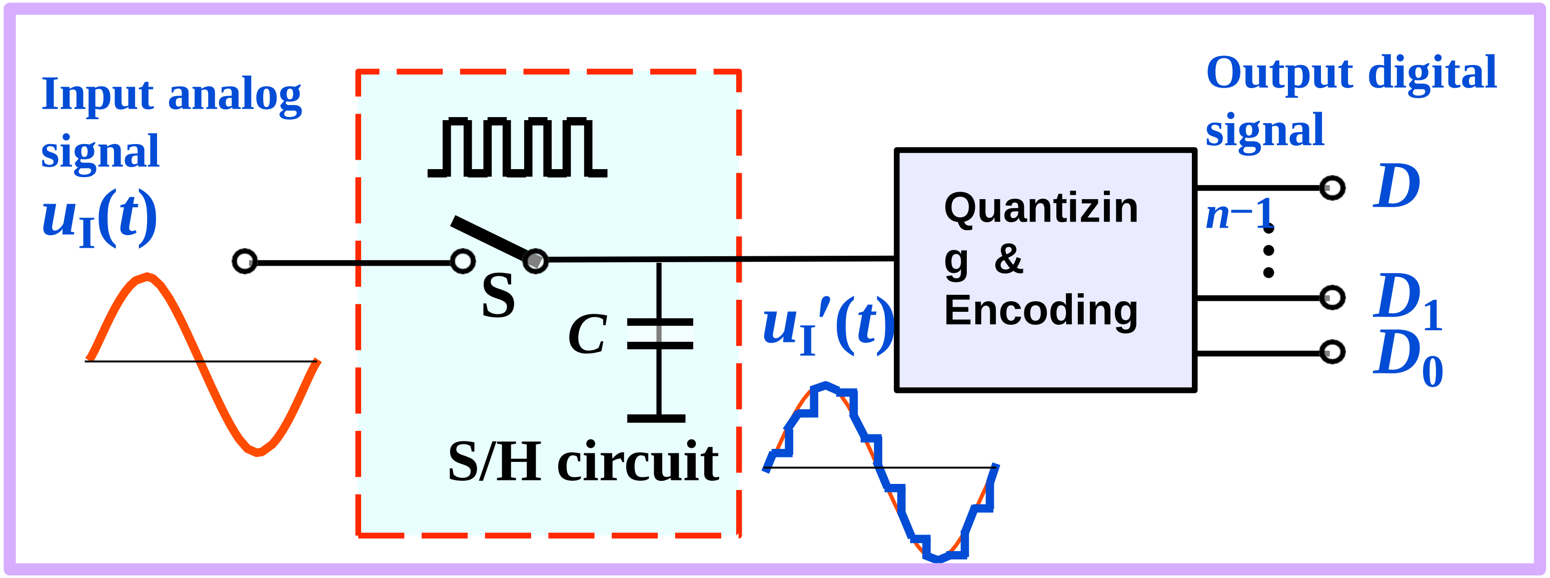
Applications of ADC

- **ADC are used virtually everywhere where an analog signal has to be processed, stored, or transported in digital form.**
- **Some examples of ADC usage are digital volt meters, cell phone, thermocouples, and digital oscilloscope.**
- **Microcontrollers commonly use 8, 10, 12, or 16 bit ADCs, our micro controller uses an 8 or 10 bit ADC.**


analog
input
ADC
sampling
quantization
binary
encoding
digital
output
1 0 1 0
Analog to Digital converter (ADC)

- https://www.youtube.com/watch?v=tZR7hLJx6Ms

ADC process

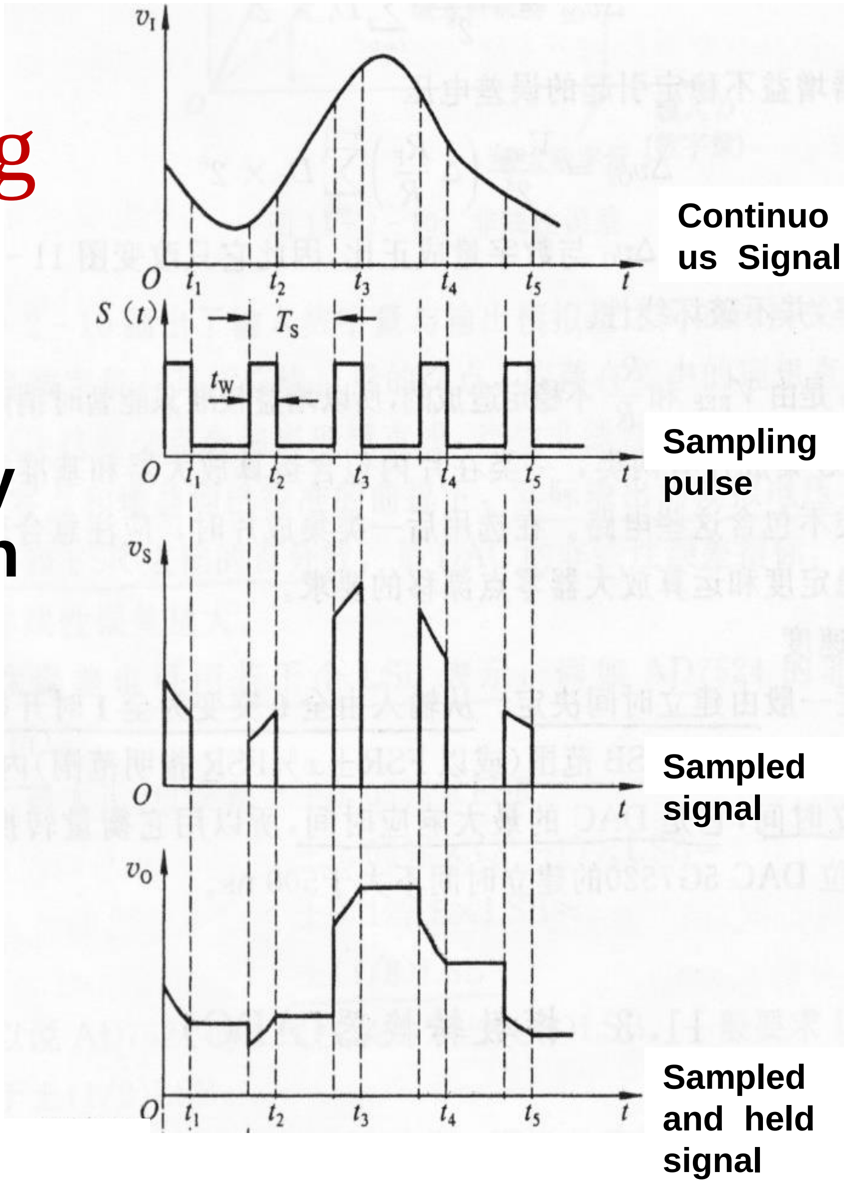


2 steps

- **Sampling and Holding (S/H)**
- **Quantizing and Encoding (Q/E)**

Sampling and Holding

- **Holding signal benefits the accuracy of the A/D conversion**
- **Minimum sampling rate should be at least twice the highest data frequency of the analog signal**



Quantizing and Encoding

Resolution:

The smallest change in analog signal that will result in a change in the digital output.

$$\Delta V = \frac{V_r}{2^N}$$

V = Reference voltage range

N = Number of bits in digital output. 2^N = Number of states.

ΔV = Resolution

The resolution represents the quantization error inherent in the conversion of the signal to digital form

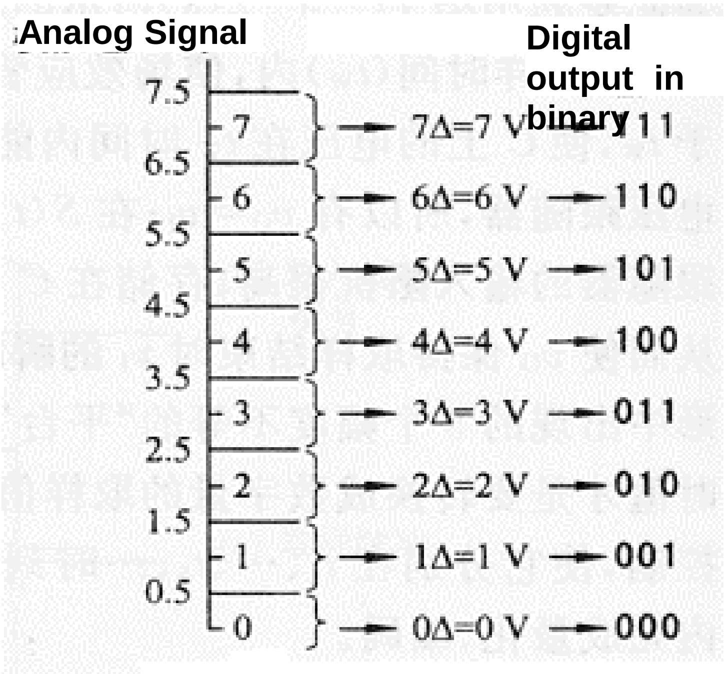
Quantizing and Encoding

- **Quantizing:**

Partitioning the reference signal range into a number of discrete quanta, then matching the input signal to the correct quantum.

- **Encoding:**

Assigning a unique digital code to each quantum, then allocating the digital code to the input signal.



!V = 1V

Ma#$m&m (&a)*$+a*$o) error $= \pm \frac{1}{2}$! $V = \pm 0.5V$

Accuracy of A/D Conversion

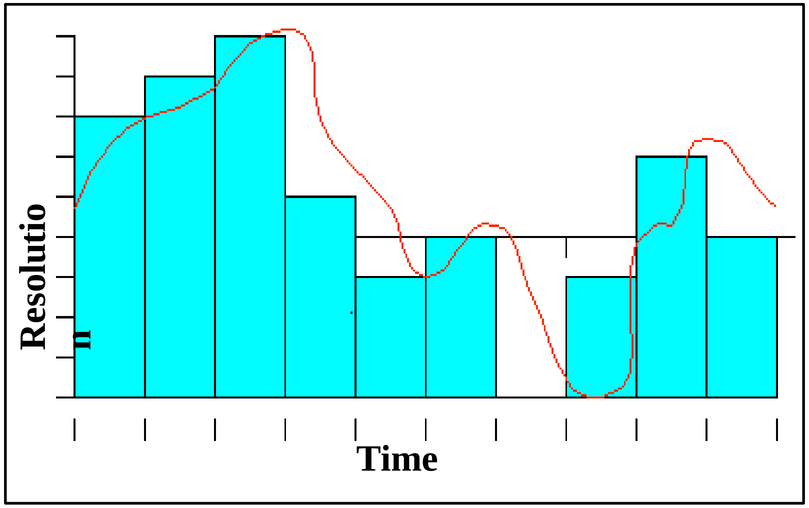
There are two ways to best improve the accuracy of A/D conversion:

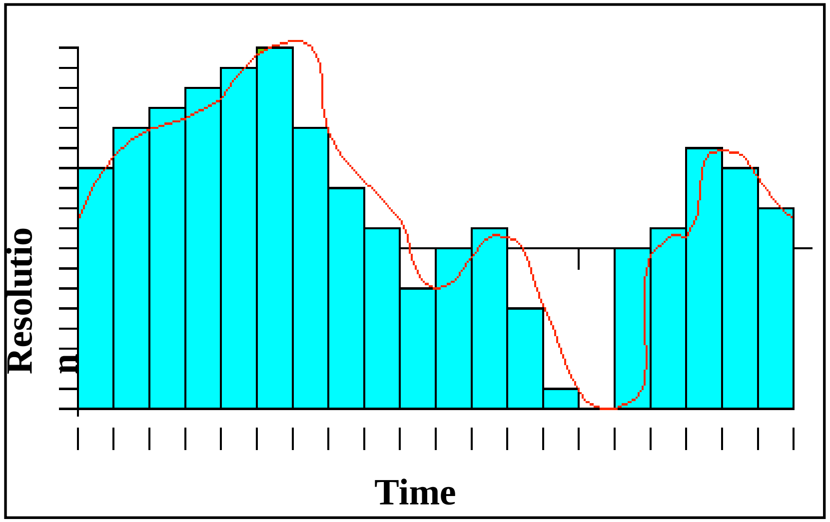
- **increasing the resolution which improves the accuracy in measuring the amplitude of the analog signal.**

- **increasing the sampling rate which increases the maximum frequency that can be measured.**

Accuracy of A/D Conversion

- **Low Accuracy**
- **Improved**



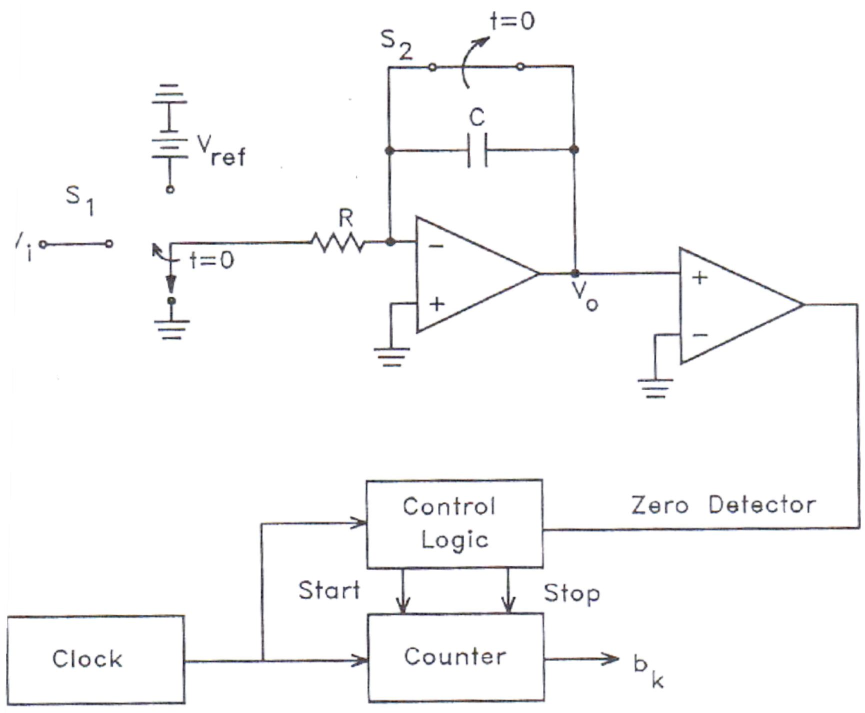


Types of A/D Converters

- **Dual Slope A/D Converter**
- **Successive Approximation A/D Converter**
- **Flash A/D Converter**
- **Delta-Sigma A/D Converter**
- **Other**
  - **Voltage-to-frequency, staircase ramp or single slope, charge balancing or redistribution, switched capacitor, tracking, and synchro or resolver**

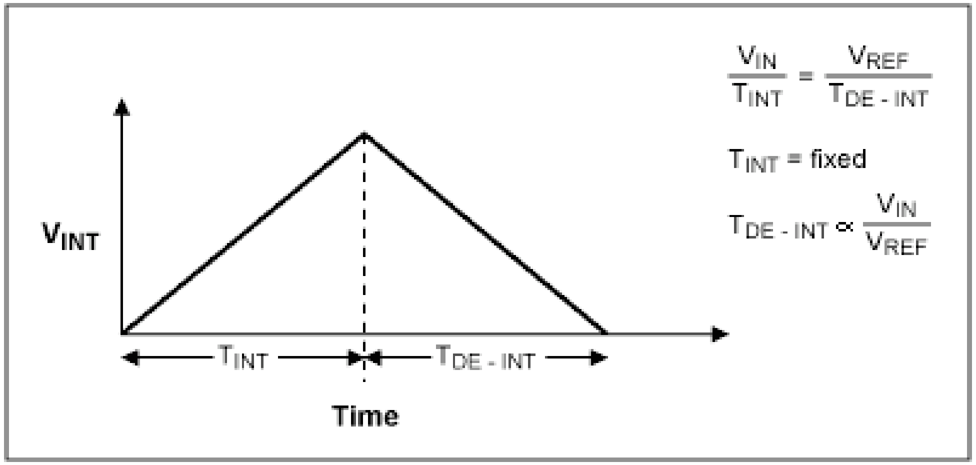
Dual Slope A/D Converter

- **Fundamental components**
  - **Integrator**
  - **Electronically Controlled Switches**
  - **Counter**
  - **Clock**
  - **Control Logic**
  - **Comparator**



How does it work

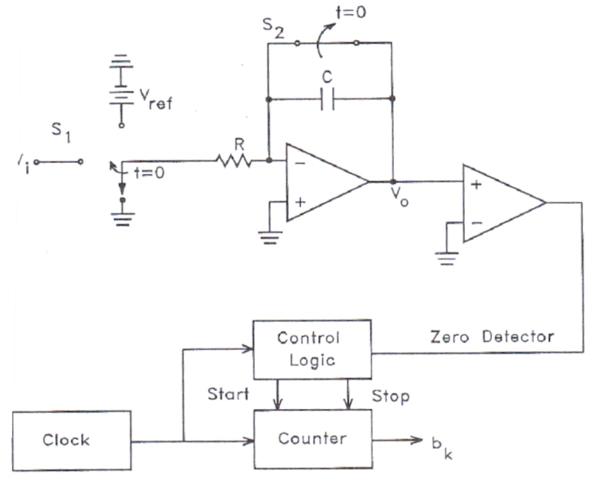
A dual-slope ADC (DS-ADC) integrates an unknown input voltage (V_{IN}) for a fixed amount of time (T_{INT}), then "de-integrates" (T_{DEINT}) using a known reference voltage (V_{REF}) for a variable amount of time.

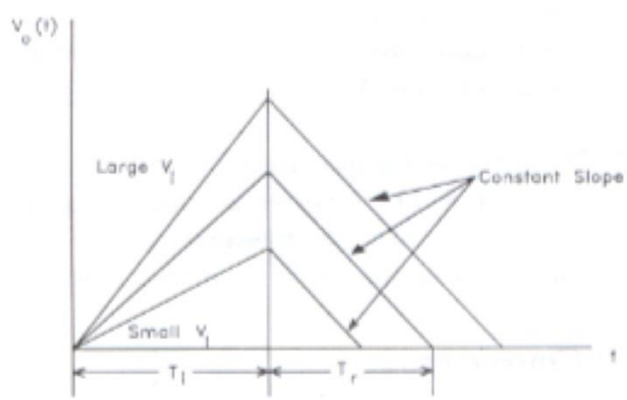


The key advantage of this architecture over the single-slope is that the final conversion result is insensitive to errors in the component values. That is, any error introduced by a component value during the integrate cycle will be cancelled out during the de-integrate phase.

How Does it Work Cont.

- At $t<0$, S_1 is set to ground, S_2 is closed, and counter=0.
- At $t=0$ a conversion begins and S_2 is open, and S_1 is set so the input to the integrator is V_{in}.
- S_1 is held for T_{INT} which is a constant predetermined time interval.
- When S_1 is set the counter begins to count clock pulses, the counter resets to zero after T_{INT}
- V_{out} of integrator at $t=T_{INT}$ is $V_{IN}T_{INT}/RC$ is linearly proportional to V_{IN}
- At $t=T_{INT}$ S_1 is set so $-V_{ref}$ is the input to the integrator which has the voltage $V_{IN}T_{INT}/RC$ stored in it.
- The integrator voltage then drops linearly with a slop $-V_{ref}/RC$.
- A compartor is used to determine when the output voltage of the integrator crosses zero
- When it is zero the digitized output value is the state of the counter.





Dual Slope A/D Converter

Pros and Cons

PROS

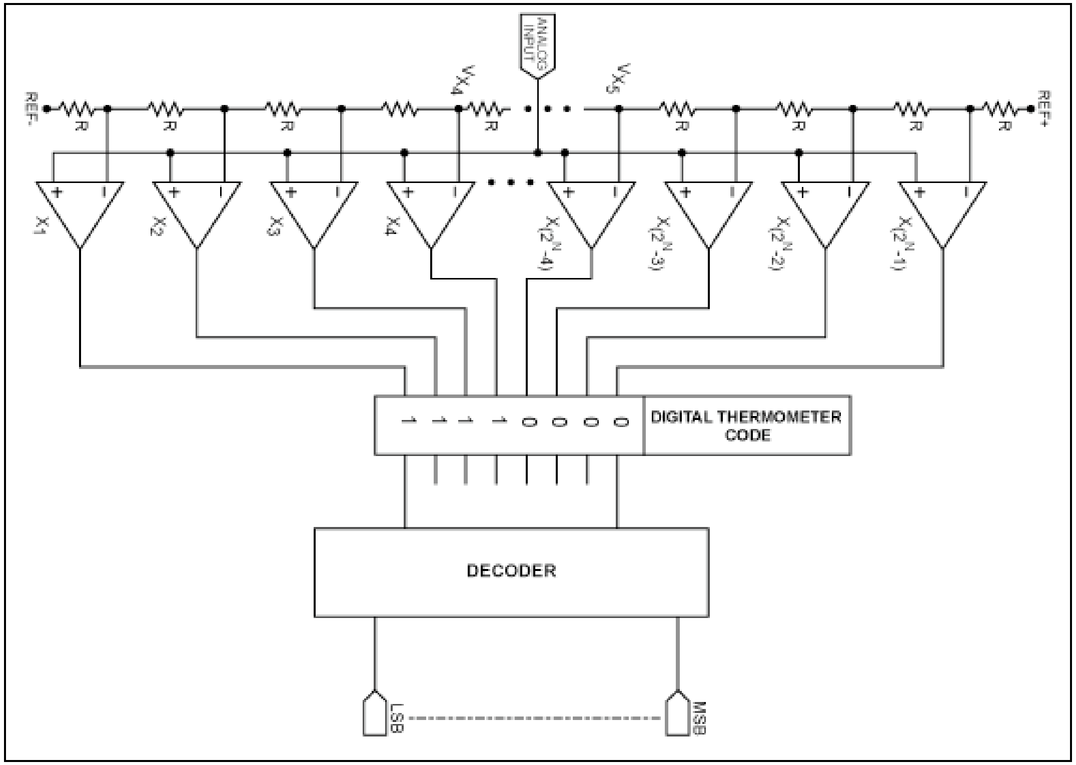
- Conversion result is insensitive to errors in the component values.
- Fewer adverse affects from “noise”
- High Accuracy

CONS

- Slow
- Accuracy is dependent on the use of precision external components
- Cost

Flash A/D Converter

- **Fundamental Components (For N bit Flash A/D)**
  - **2^N-1 Comparators**
  - **2^N Resistors**
  - **Control Logic**



How does it work

- **Uses the 2^N resistors to form a ladder voltage divider, which divides the reference voltage into 2^N equal intervals.**
- **Uses the 2^N-1 comparators to determine in which of these 2^N voltage intervals the input voltage V_{in} lies.**
- **The Combinational logic then translates the information provided by the output of the comparators**
- **This ADC does not require a clock so the conversion time is essentially set by the settling time of the comparators and the propagation time of the combinational logic.**

Flash A/D Converter

Pros and Cons

PROS

- **Very Fast (Fastest)**
- **Very simple operational theory**
- **Speed is only limited by gate and comparator propagation delay**

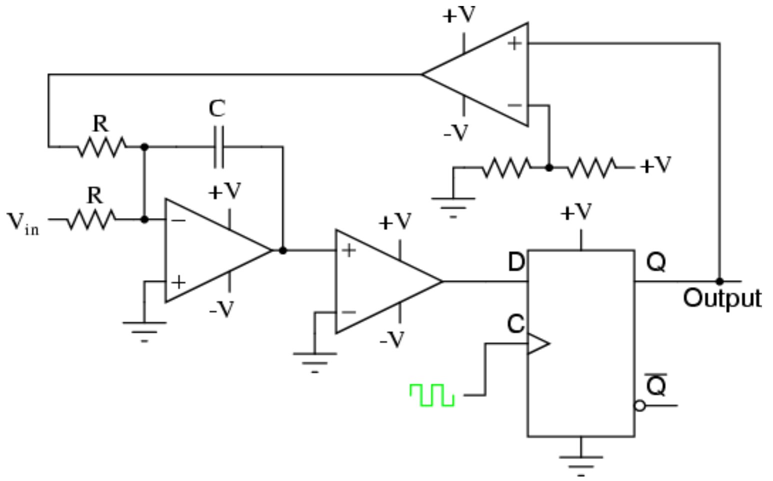
CONS

- **Expensive**
- **Prone to produce glitches in the output**
- **Each additional bit of resolution requires twice the comparators.**

SIGMA-DELTA A/D Converter

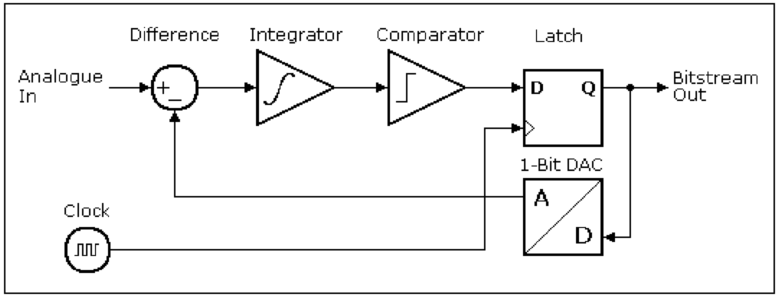
Main Components

- **Resistors**
- **Capacitor**
- **Comparators**
- **Control Logic**
- **DAC**



How does it work

- Input is over sampled, and goes to integrator.
- The integration is then compared to ground.
- Iterates and produces a serial bit stream
- Output is a serial bit stream with # of 1's proportional to V_{in}



- With this arrangement the sigma-delta modulator automatically adjusts its output to ensure that the average error at the quantizer output is zero.
- The integrator value is the sum of all past values of the error, so whenever there is a non-zero error value the integrator value just keeps building until the error is once again forced to zero.

Sigma-Delta A/D Converter

Pros and Cons

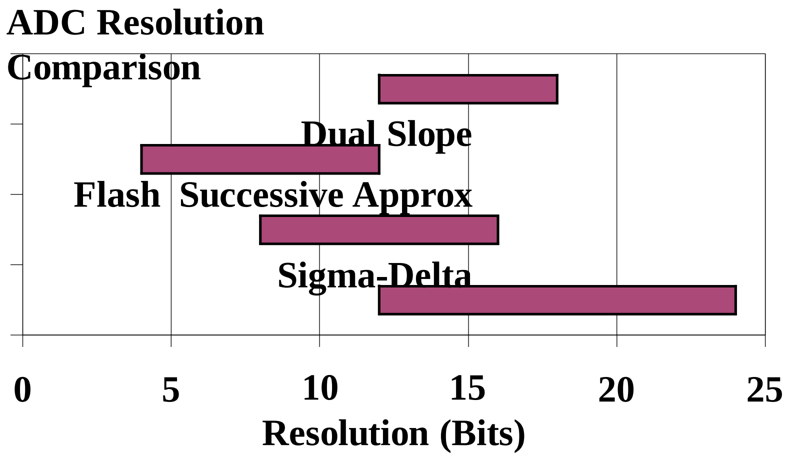
PROS

- **High Resolution**
- **No need for precision components**

CONS

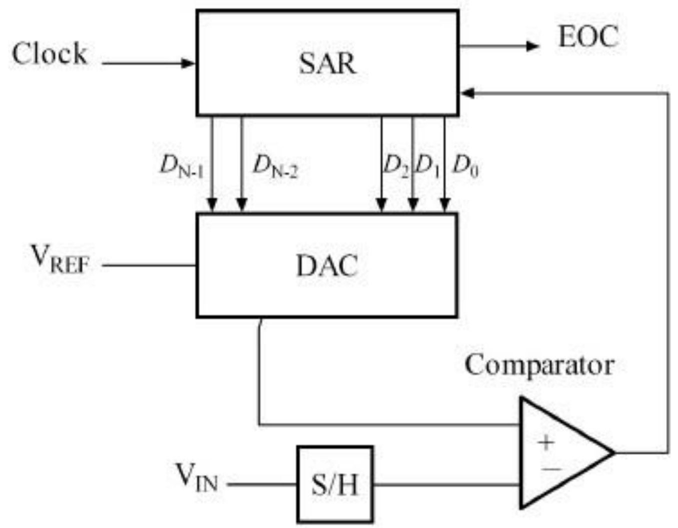
- **Slow due to over sampling**
- **Only good for low bandwidth**

ADC Types Comparison



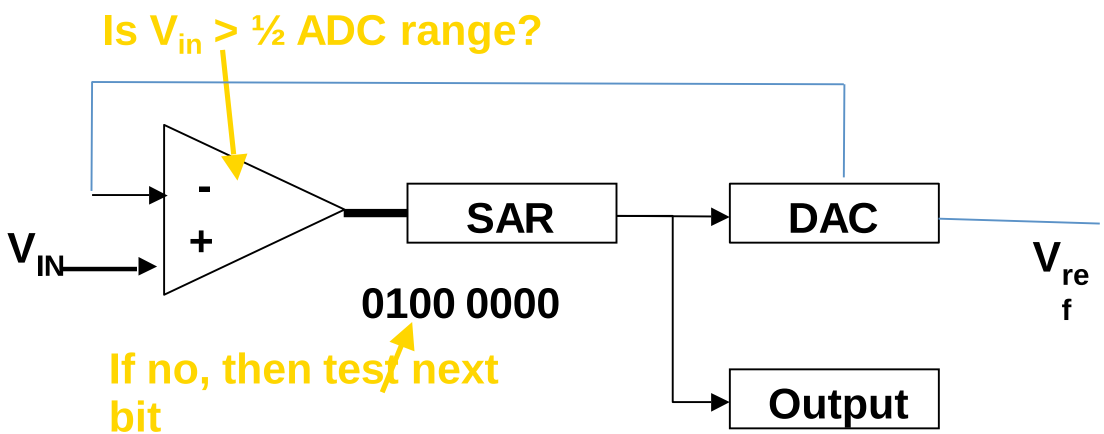
Type	Speed (relative)	Cost (relative)
Dual Slope	Slow	Med
Flash	Very Fast	High
Successive Appox	Medium – Fast	Low
Sigma-Delta	Slow	Low

Successive Approximation ADC Circuit



- **Uses a n-bit DAC to compare DAC and original analog results.**
- **Uses Successive Approximation Register (SAR) supplies an approximate digital code to DAC of Vin.**
- **Comparison changes digital output to bring it closer to the input value.**
- **Uses Closed-Loop Feedback Conversion**

Successive Approximation ADC

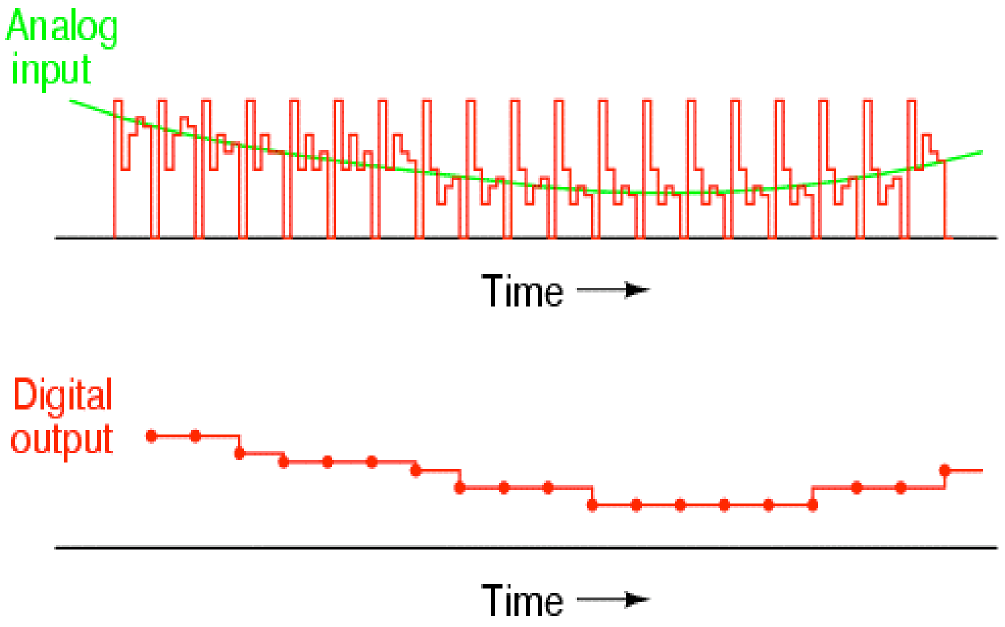


Process

1. **MSB initialized as 1**
2. **Convert digital value to analog using DAC**
3. **Compares guess to analog input**
4. **Is $V_{in}>V_{DAC}$**
    - **Set bit 1**
    - **If no, bit is 0 and test next bit**

Closed-Loop

Output



Successive Approximation

Advantages

- Capable of high speed and reliable
- Medium accuracy compared to other ADC types
- Good tradeoff between speed and cost
- Capable of outputting the binary number in serial (one bit at a time) format.

Disadvantages

- Higher resolution successive approximation ADC's will be slower
- Speed limited to ~5Msps

Successive Approximation Example

Example

- **10 bit ADC**
- **Vin= 0.6 volts (from analog device)**
- **V_{ref}=1 volts**
- **Find the digital value of Vin**

Bit	Voltage
9	.5
8	.25
7	.125
6	.0625
5	.03125
4	.015625
3	.0078125
2	.00390625
1	.001952125
0	.0009765625

$N=2^n$ (N of possible states)
N=1024
Vmax-Vmin/N = 1 Volt/1024 = 0.0009765625V of Vref (resolution)

Successive Approximation

- **MSB (bit 9)**
  - **Divided V_{ref} by 2**
  - **Compare V_{ref} /2 with V_{in}**
  - **If V_{in} is greater than V_{ref} /2 , turn MSB on (1)**
  - **If V_{in} is less than V_{ref} /2 , turn MSB off (0)**
  - **V_{in} =0.6V and V=0.5**
  - **Since V_{in}>V, MSB = 1 (on)**

Bit	Voltage
9	.5
8	.25
7	.125
6	.0625
5	.03125
4	.015625
3	.0078125
2	.00390625
1	.001952125
0	.0009765625

1									

Successive Approximation

- **Next Calculate MSB-1 (bit 8)**
  - **Compare V_{in}=0.6 V to V=V_{ref}/2 + V_{ref}/4= 0.5+0.25 =0.75V**
  - **Since 0.6<0.75, MSB is turned off**
- **Calculate MSB-2 (bit 7)**
  - **Go back to the last voltage that caused it to be turned on (Bit 9) and add it to V_{ref}/8, and compare with V_{in}**
  - **Compare V_{in} with (0.5+V_{ref}/8)=0.625**
  - **Since 0.6<0.625, MSB is turned off**

1	0	0							

Successive Approximation

- **Calculate the state of MSB-3 (bit 6)**
  - **Go to the last bit that caused it to be turned on (In this case MSB-1) and add it to V_{ref}/16, and compare it to**

V_{in}

  - **Compare V_{in} to V= 0.5 + V_{ref}/16= 0.5625**
  - **Since 0.6>0.5625, MSB-3=1 (turned on)**

MSB	MSB-1	MSB-2	MSB-3	…					
1	0	0	1						

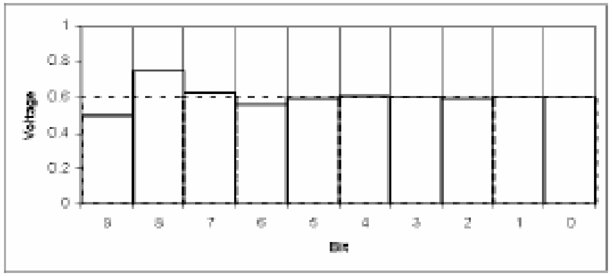
Successive Approximation ADC

- **This process continues for all the remaining bits.**

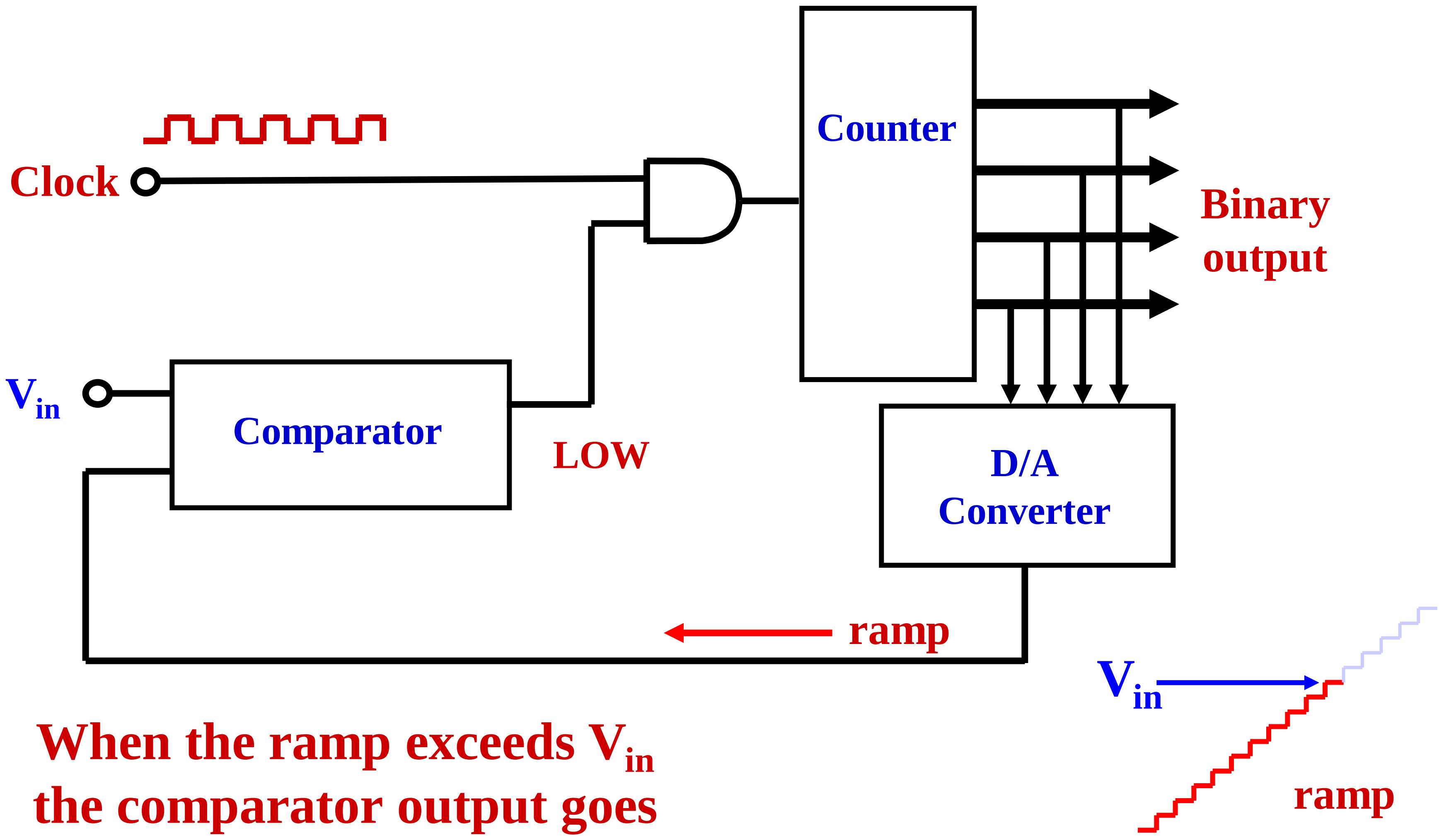
•Digital Results:

MSB	MSB-1	MSB-2	MSB-3	…					LSB
1	0	0	1	1	0	0	1	1	0

•Results: $$\frac{1}{2}+\frac{1}{16}+\frac{1}{32}+\frac{1}{256}+\frac{1}{512}=.599609375\text{ V}$$

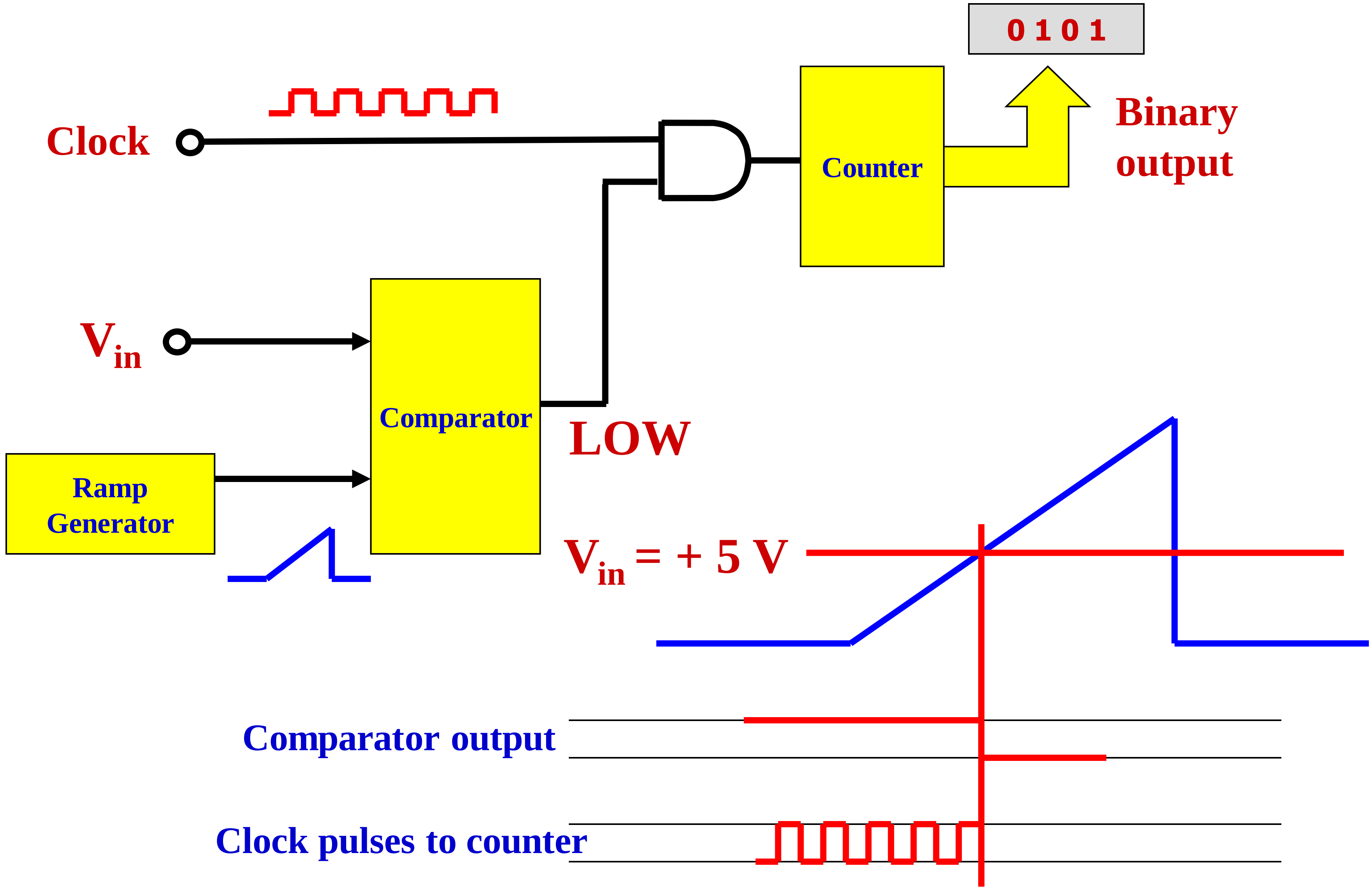


COUNTER-RAMP TYPE A/D CONVERTER

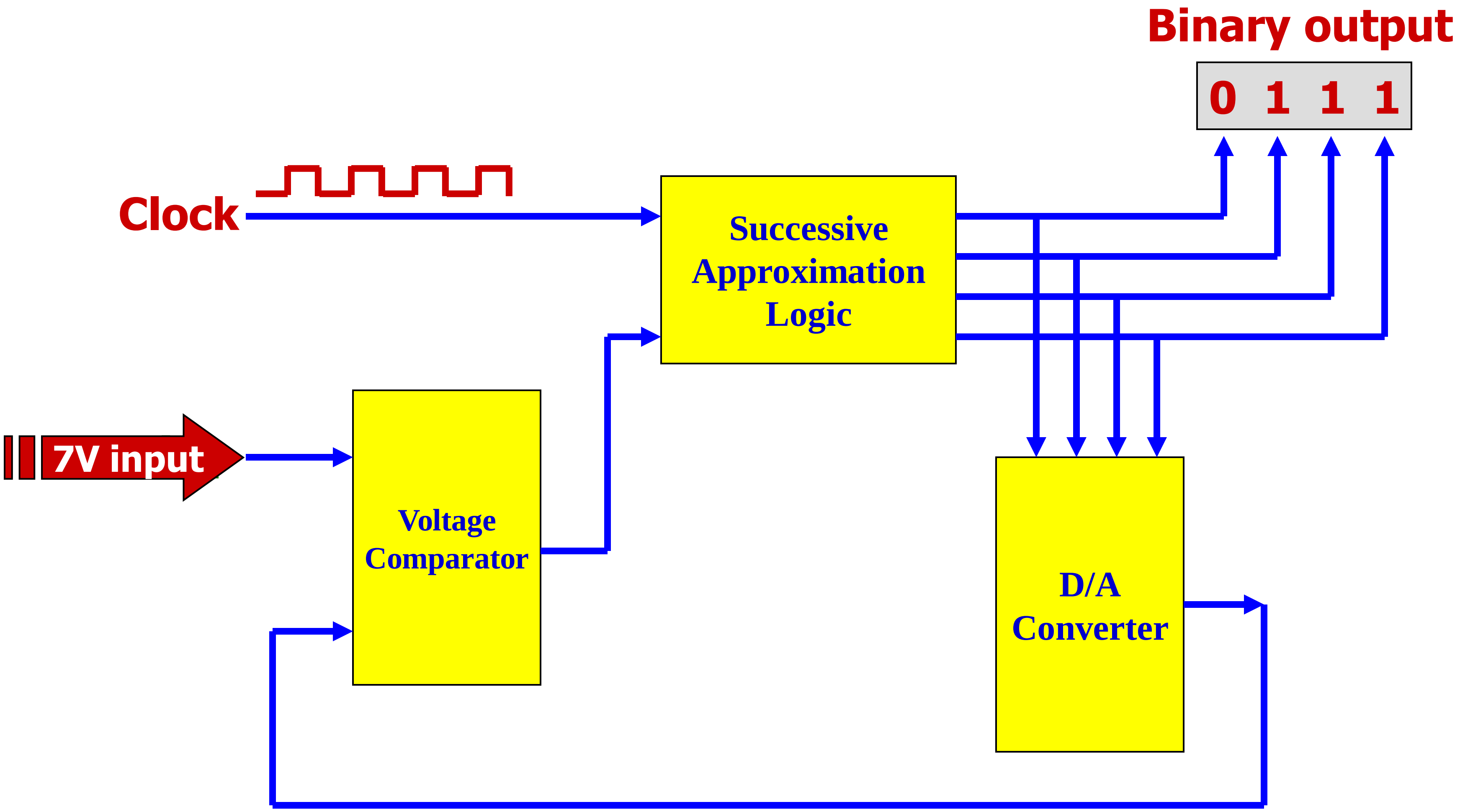


When the ramp exceeds V_{in} the comparator output goes LOW and the clock is disabled.

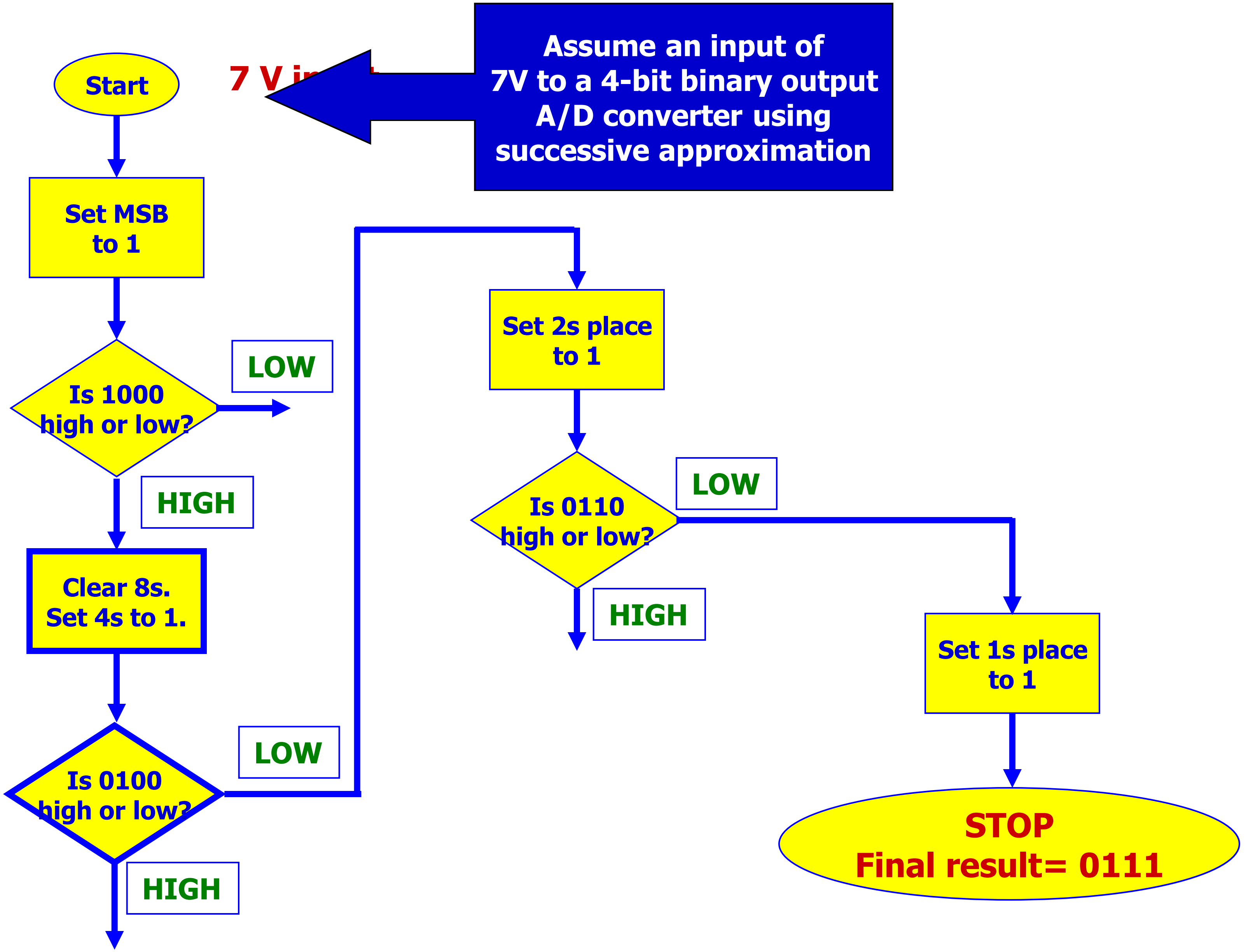
RAMP-TYPE A/D CONVERTER



SUCCESSIVE-APPROXIMATION TYPE A/D CONVERTER



FLOWCHARTING SUCCESSIVE APPROXIMATION



A/D CONVERTER SPECIFICATIONS

- Type of output
  - Binary (microprocessor-type)
  - Decimal (digital meter type)

- Resolution
  - Binary output- number of bits (such as 4, 6, 8, 12, 14 or 16)
  - Decimal output- number of digits on display (such as 3 1/2 or 4 1/2)

- Accuracy
  - Binary output- (ranges from +/- 1/2 LSB to about +/- 2 LSB)
  - Decimal output- (ranges from about 0.01 to 0.05 percent)

- Conversion time is the time it takes for the IC to convert the analog input voltage to digital output. Binary output A/D converters commonly have faster conversion times than decimal output types.